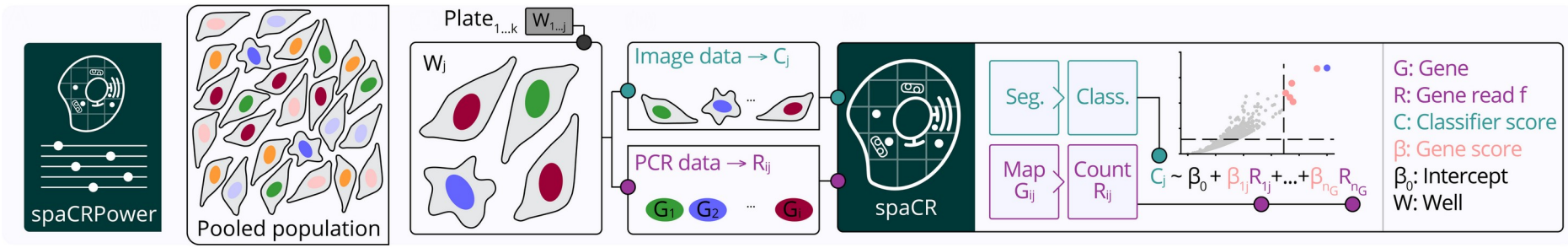
# spaCR

## Spatial phenotype analysis of CRISPR screens

spaCR 1.5.1.4 · github.com/EinarOlafsson/spacr

INTRODUCTION

# What is spaCR?




*Pooled screen: every well holds a mix of knockouts. Image it, sequence it, and regress. (Preprint, Fig. 1A)*

## Image every well

Segment, measure and classify every cell. The phenotype becomes a number per well.

## Sequence every well

Read the gRNA barcodes to learn which knockouts are in each well, and in what proportion.

## Link the two

Regress the phenotype on guide composition. Out comes an effect per gene: a ranked hit list.

**No sequencing?** Segmentation, measurement, annotation, classification and the assays work on any high-content imaging experiment.

INTRODUCTION

# Where spaCR fits among existing tools

tool lacks it
tool has it: spaCR has it too
tool has it: spaCR does not

| | CellProfiler | Fiji / ImageJ | napari | Cellpose | ilastik | QuPath | DL4Mic | MAGeCK | spaCR | Category |
|---|---|---|---|---|---|---|---|---|---|---|
| *Best for* | *cell measurement* | *image analysis* | *image viewing* | *segmentation* | *pixel / object ML* | *tissue slides* | *no-code DL* | *sequencing screens* | *image-based screens* | |
| Point-and-click interface | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | limited | ✗ | ✓ | Usability |
| Batch and headless runs (server, cluster) | ✓ | limited | ✗ | ✓ | ✓ | ✓ | limited | ✓ | ✓ | Automation |
| Reads most microscope file formats | ✓ | ✓ | plugin | limited | limited | ✓ | limited | ✗ | ✓ | Input |
| Deep-learning segmentation | plugin | plugin | plugin | ✓ | limited | plugin | ✓ | ✗ | ✓ | Segmentation |
| Train your own segmentation model | ✗ | plugin | plugin | ✓ | ✓ | limited | ✓ | ✗ | ✓ | Segmentation |
| Live assisted segmentation | ✗ | plugin | plugin | limited | ✓ | ✓ | ✗ | ✗ | ✓ | Segmentation |
| Correct masks by hand | limited | ✓ | ✓ | ✓ | ✓ | ✓ | ✗ | ✗ | ✓ | Segmentation |
| Per-cell measurements | ✓ | ✓ | plugin | ✗ | ✓ | ✓ | ✗ | ✗ | ✓ | Measurement |
| 3-D (z-stack) images | ✓ | ✓ | ✓ | ✓ | ✓ | limited | limited | ✗ | **beta** | Image types |
| Time-lapse tracking | ✓ | plugin | plugin | ✗ | ✓ | ✗ | ✗ | ✗ | ✓ | Image types |
| Whole plates in one run | ✓ | limited | ✗ | limited | limited | limited | limited | ✗ | ✓ | Screening |
| Plate maps, heat maps and QC | limited | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ | Screening |
| Classify cells from images (deep learning) | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ | Classification |
| Classify cells from measurements | limited | ✗ | plugin | ✗ | ✓ | ✓ | ✗ | ✗ | ✓ | Classification |
| Explore cells: UMAP with images, gating | limited | ✗ | plugin | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ | Analysis |
| Plots and statistics without code | limited | limited | plugin | ✗ | ✗ | limited | ✗ | limited | ✓ | Analysis |
| GPU acceleration | ✗ | plugin | plugin | limited | limited | limited | ✓ | ✗ | ✓ | Performance |
| Pooled CRISPR screens: images to genes | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | limited | ✓ | Screening |
| Pixel classifier from brush strokes | ✗ | plugin | plugin | ✗ | ✓ | ✓ | ✗ | ✗ | ✗ | Segmentation |
| Whole-slide / tissue images | ✗ | limited | ✗ | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ | Image types |
| Large plugin ecosystem | limited | ✓ | ✓ | ✗ | ✗ | limited | ✗ | ✗ | ✗ | Extensibility |

*Core features; 'plugin' = needs an add-on, 'limited' = partly. Simplified: check each tool for details.*

INTRODUCTION

ALPHA

# Alpha features: switch them on in Preferences

| Category | Alpha feature |
|---|---|
| Segmentation | StarDist segmentation backend |
| | InstanSeg segmentation backend |
| | Omnipose segmentation backend |
| | DINOCell segmentation backend |
| | SAMCell segmentation backend |
| | Spotiflow spot detection |
| | SpotNet spot detection |
| | Click- and box-prompted segmentation (micro-SAM) |
| | Segmentation uncertainty maps |
| | Parameter-robustness report |
| | Run CellProfiler .cppipe pipelines |
| | Masks and ROIs in and out: QuPath, ImageJ, COCO |
| | Virtual staining from brightfield |
| | Mask batches across several GPUs |
| | Use Make Masks settings in Mask Generation |
| | MitoNet: mitochondria in TEM images |
| Image preprocessing | Spectral unmixing (bleed-through correction) |
| | Photobleaching correction |
| | Vendor flat-field profiles (Harmony, ZEN) |
| | Intensity calibration across sessions |
| | Deep-learning image QC: focus, debris, empty fields |
| | Self-supervised denoising (Noise2Void), low-light benchmark |
| | Illumination, enhancement and PSF settings |
| Measurements and assays | Cell-cycle phase classification |
| | Live/dead viability and cytotoxicity index |
| | Confluency for any channel |
| | Scratch / wound-closure assay |
| | Colony / CFU counting |
| | Image-based profiling: normalise, select, mAP |
| | GPU measurement (cuCIM, CuPy) |
| | Estimate plaque scale and growth time |

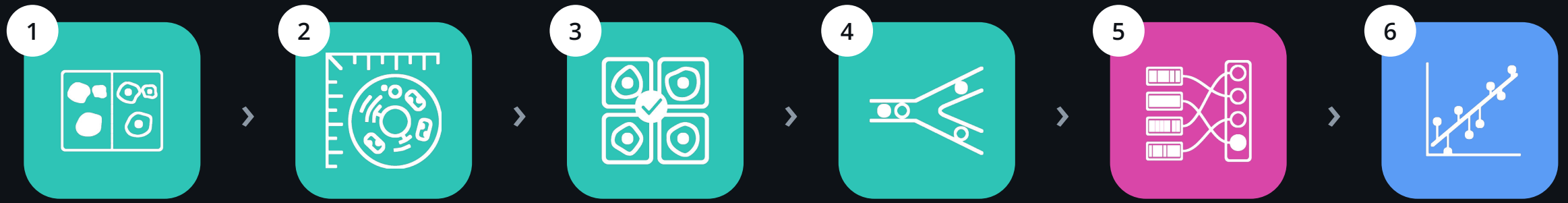
| Category | Alpha feature |
|---|---|
| Time-lapse | SAM2 propagation: segment once, follow each object |
| | Timeflows: a temporal Cellpose |
| | Lineage trees from tracks |
| | Event detection: mitosis, egress, invasion, death |
| | Time-to-event: Kaplan-Meier and Cox |
| Cells and machine learning | Foundation-model embeddings (SubCell, OpenPhenom, …) |
| | Well labels to responding cells (multiple-instance) |
| | Anomaly detection against negative controls |
| | Find cells like this one (similarity search) |
| | Blind scoring |
| | Counterfactuals: morph a control cell toward a hit |
| | Self-supervised DINO pretraining |
| | Judge every label from cross-channel ground truth |
| Screens and statistics | Arrayed-screen hits: SSMD, robust z, B-score |
| | Chemistry-aware hits: SMILES, clusters, SAR |
| | Sample-size planner for arrayed assays |
| | Spatial transcriptomics: Visium, Xenium; spatial masks, barcodes |
| | LIMS / plate-barcode linkage |
| Automation and data | Watch a folder, analyse images as they arrive |
| | Smart microscopy: send hits back for re-imaging |
| | Cloud storage (S3 and others) |
| | Export a run as a Nextflow / Snakemake workflow |
| | Run-finished notifications (email, Slack, ntfy) |
| | Export for R (SingleCellExperiment), tidy Parquet |
| | Plugin and recipe catalogue |
| | Postgres or DuckDB / Parquet measurement store |
| | Test data on seven more modules |
| | Export masks and objects for vvvv |
| Reproducibility | Figure-integrity check |
| | Pre-registered analysis lock |
| | Metadata and archive packages (REMBI, IDR) |
| | One-click Zenodo archive with a DOI |

| Alpha species | Plasmodium | Candida | Trypanosoma | Leishmania | Giardia | Virus | Mammalian |
|---|---|---|---|---|---|---|---|

63 alpha features and 7 alpha species, hidden until Preferences → Modules → Show alpha features and Show alpha species are on. Early versions: expect changes.

INTRODUCTION

# Six steps from plate to hit list




### Mask
Find every cell, nucleus, parasite

### Measure
Shape, intensity, texture per object

### Annotate
Label example cells

### Classify
Score every cell's phenotype

### Map Barcodes
Which gRNAs are in which well

### Regression
Which genes change the phenotype

images

sequencing + statistics

**One project, one database.** Images, masks, crops, measurements, annotations, predictions, barcodes and well IDs all live in one SQLite file (measurements.db). Every module reads from it and writes back to it, and every run saves its settings so it can be replayed.

INTRODUCTION

# 19 modules in one desktop app

*The home screen: every module is a tile, grouped by what it is for*

**6** — **Core**
Mask · Measure · Annotate · Classify · Map Barcodes · Regression

**7** — **Data**
Import · Embeddings · Run Compare · Experiment Design · Power · Dose–Response · QC

**5** — **Tools**
Make Masks · Align & Stitch · Image UMAP · Gate Editor · Graph Builder

**1** — **Assays**
Toxoplasma compartments and assays; other organisms appear when alpha species are enabled.

Plus ~40 sub-tools inside the modules. Ctrl+K opens a command palette that reaches every one of them.

INTRODUCTION

# Runs on a laptop, a workstation or a cluster

## Desktop app

Point-and-click Qt interface. Live previews, a console per module, settings search, recipes, and a Run button.

## Headless

The same modules and settings files run without the window, on a server or a cluster: or send jobs there from the app.

## Reproducible

Every run records its settings. spacr-repro RUN_DIR replays it. Run History and Run Compare show what changed.

**4**

**kinds of graphics card**

NVIDIA, AMD, Apple and Intel: or just the CPU

**10**

**interface languages**

switch in Preferences without restarting

**84**

**interactive tutorials**

narrated videos from install to hit investigation

**3.9–3.14**

**Python versions**

3.12 recommended; Linux, macOS, Windows

INTRODUCTION

# Install spaCR

## Desktop installer

*Easiest. Bundles its own Python: no conda needed.*

```
GitHub → Releases → latest
Windows   .exe
macOS     .pkg  (Intel + Apple silicon)
Linux     chmod +x SpaCR-*.run && ./SpaCR-*.run
```

## pip (conda env)

*Most flexible; add optional extras.*

```
conda create -n spacr python=3.12 -y
conda activate spacr
pip install "spacr[qt]"
spacr
```

## conda-forge

*One command into an existing env.*

```
conda create -n spacr python=3.12 -y
conda activate spacr
conda install conda-forge::spacr
spacr
```

## From source

*For development (nightly branch).*

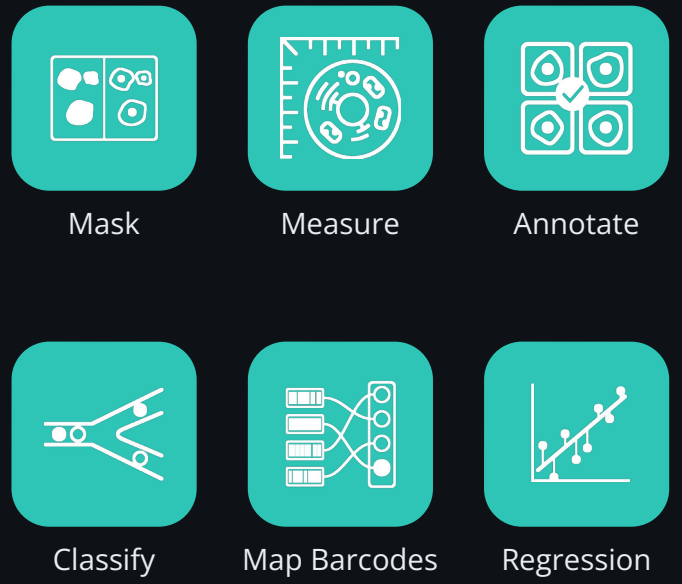
```
git clone --branch nightly \
  https://github.com/EinarOlafsson/spacr
cd spacr && pip install -e .
spacr
```

**Something off?** `spacr-doctor` checks the install and says how to fix it. Headless server: `pip install spacr` (no Qt).

# 01

# Core pipeline

From raw plate images and FASTQ reads to a ranked list of genes.



CORE · 1 OF 6

# Mask: segmentation

STABLE

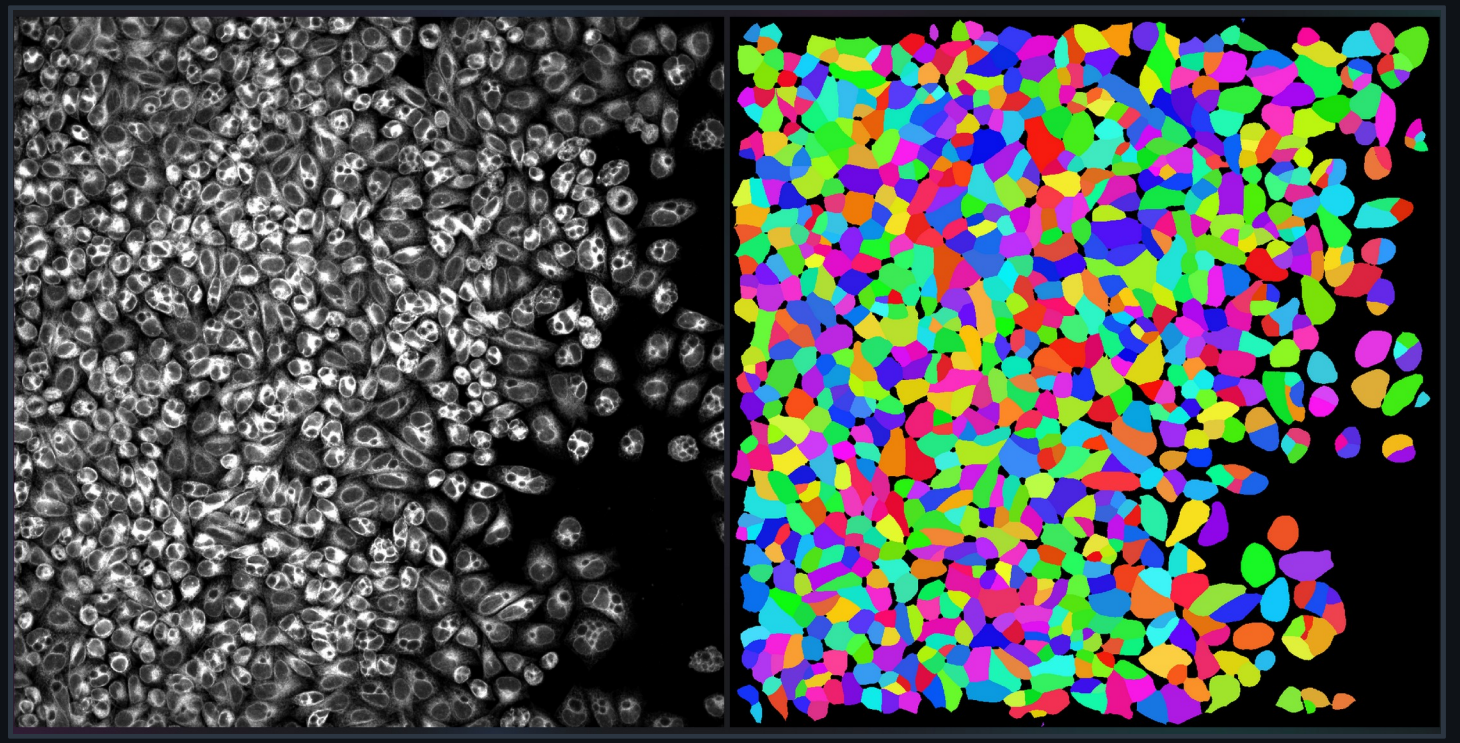
*Finds and outlines every cell, nucleus, parasite and organelle in your images.*

## WHAT YOU CAN DO

- Reads files from most microscopes (TIFF, Leica, Zeiss, Nikon)
- Cellpose-SAM by default; other segmentation models optional
- Cell, nucleus, pathogen and up to 26 organelle classes
- Plug in fine-tuned Model Zoo models (e.g. Toxoplasma vacuoles)
- 2-D, 3-D z-stacks and time-lapse; live preview before a full run

## OUTPUT

- One mask per object type, aligned to the images
- Overlay images and a quality score per field
- The settings used, saved with the run

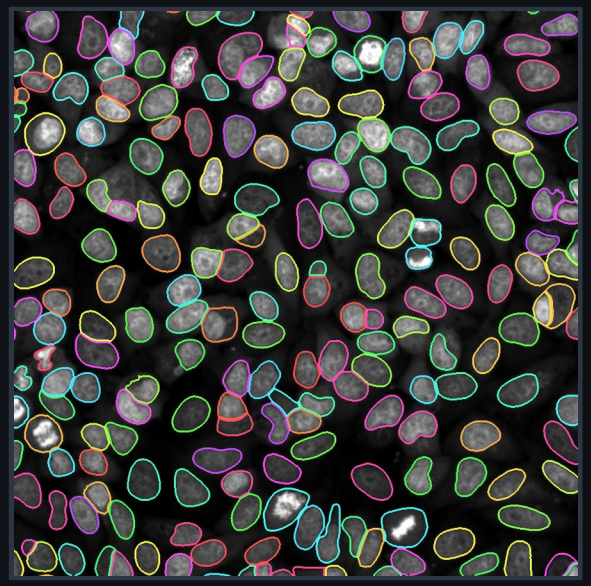


*Live preview: host cells (left) and the label mask Cellpose-SAM returns (right)*

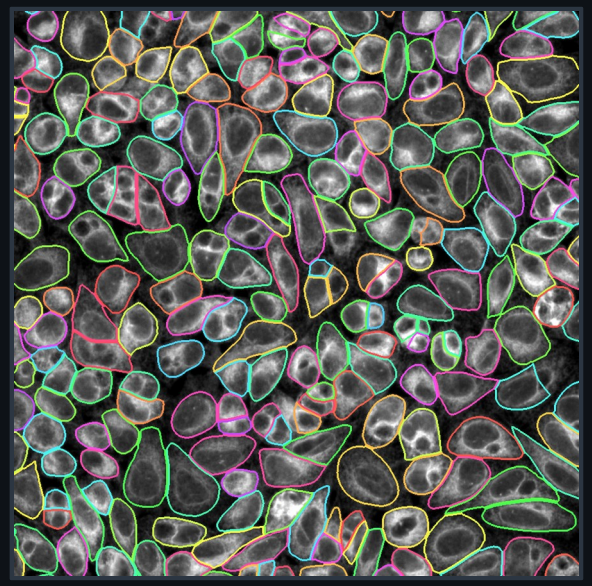
CORE · 1 OF 6

# Mask: every object class

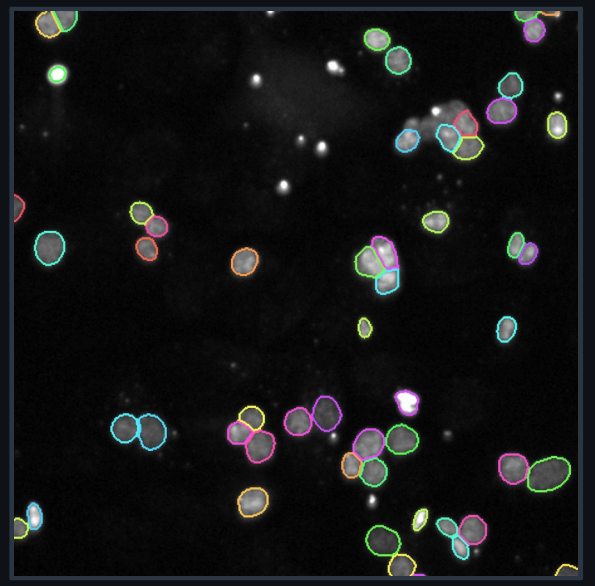
STABLE



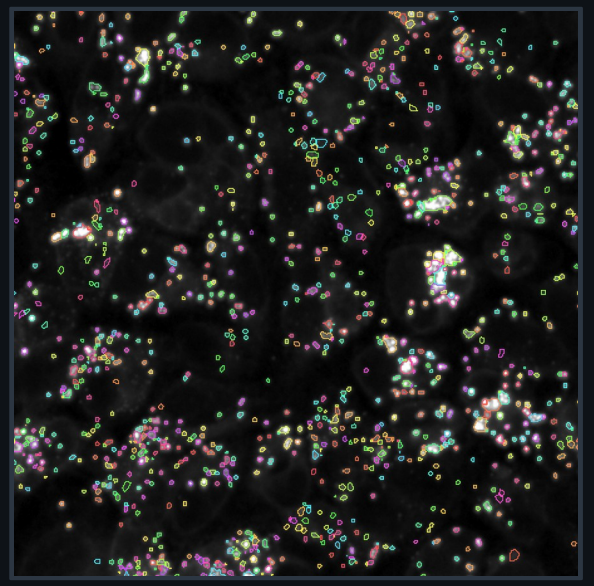
*Nuclei: Cellpose-SAM (Hoechst)*



*Cells: Cellpose-SAM (cell stain)*



*Pathogen: Toxoplasma, zoomed*



*Organelles: lipid droplets, zoomed*

## SUB-TOOLS INSIDE THIS MODULE

**Timelapse**
Follow each cell and parasite from frame to frame in movies

**Cellpose flows**
See why two cells were merged, or one was split

CORE · 2 OF 6

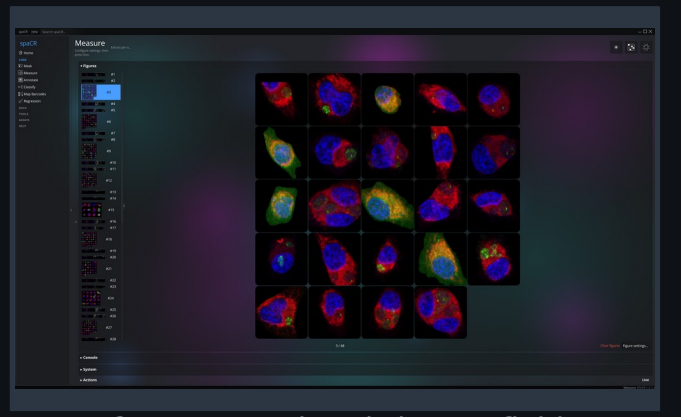
# Measure: features and crops

STABLE

*Turns every segmented object into a row of numbers: shape, brightness, texture and position.*

## WHAT YOU CAN DO

- Shape, size, brightness and texture of every object
- Position: neighbours, distances, touching the edge
- Channel overlap; measure inside a chosen region
- Saves object crops for Annotate and Classify
- 3-D objects measured in µm
- Checks free RAM first and sizes its workers to fit



*Object crops from a completed sixteen-field Measure run*

*The cell table in the Database Browser*

## OUTPUT

- One table per object type (cells, nuclei, parasites, …)
- Per-object PNG crops

## SUB-TOOLS INSIDE THIS MODULE

**Illumination**
Flat-field correction estimated from the plate itself

**AnnData export**
Export for single-cell analysis tools (scanpy)

**Motility assay**
Track velocity and straightness, split by infection state

**Database Browser**
Browse, filter and export any project table

CORE · 3 OF 6

# Annotate: label cells by hand

STABLE

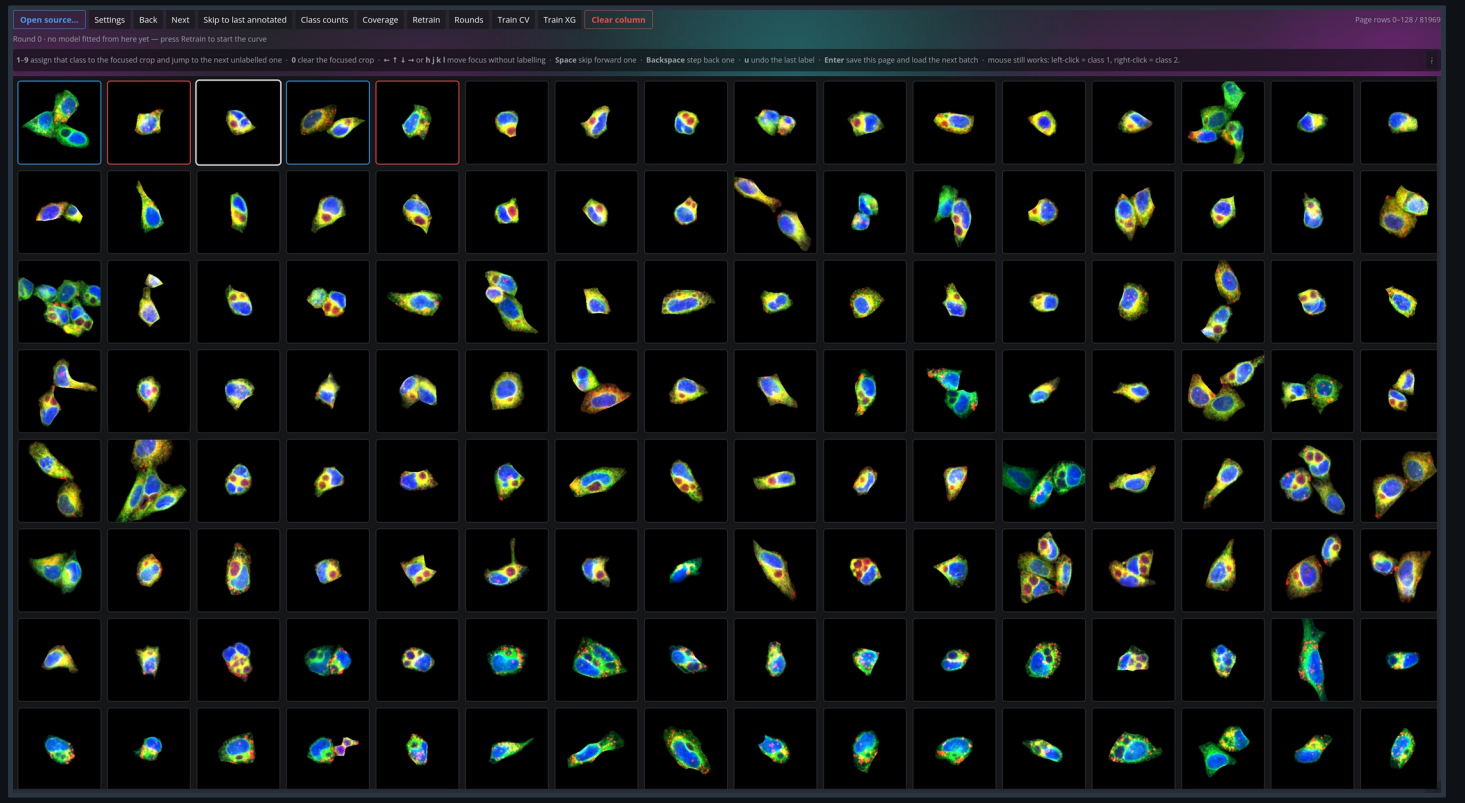
*Label single-cell crops so a classifier can learn what each phenotype looks like.*

## WHAT YOU CAN DO

- Pages through a grid of crops from the project
- Click, or press 1–9 with auto-advance; 0 clears, u undoes
- Several annotation columns side by side (e.g. two annotators)
- Active learning: shows the cells the model is least sure about first
- Change channels and colours for display only; files stay untouched

## OUTPUT

- Labels in an annotation column of the project
- Class counts and coverage per well




*Annotation grid. Selected crops are outlined; the queue can be ranked by model uncertainty*

## SUB-TOOLS INSIDE THIS MODULE

**Annotator Agreement**
How often two people label a cell the same way: and a list of the cells they disagreed on

CORE · 4 OF 6

# Classify: score every cell

ALPHA

*Teach the computer what a phenotype looks like, then let it score every cell in the screen.*

## WHAT YOU CAN DO

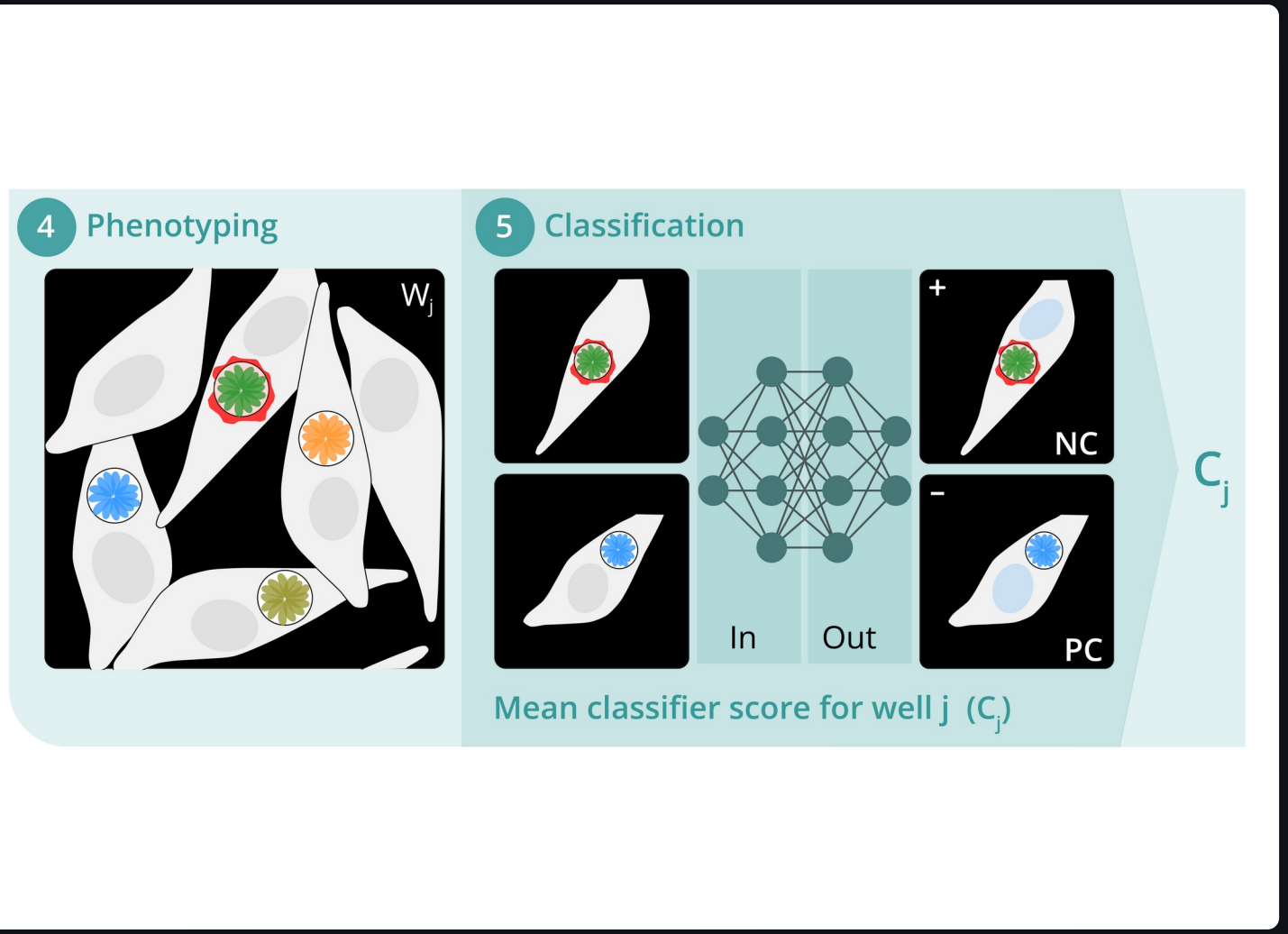
**Image model:** learns straight from the cell pictures (ResNet, EfficientNet, ViT, MaxViT, …)

**Measurement model:** learns from the measured features (XGBoost, random forest, …)

- Examples come from control wells or from your Annotate labels
- Tested on wells it never trained on, so the accuracy is honest
- Scores every cell and writes the result back to the project

## OUTPUT

- A class and a probability for every cell
- The mean score per well → input to Regression

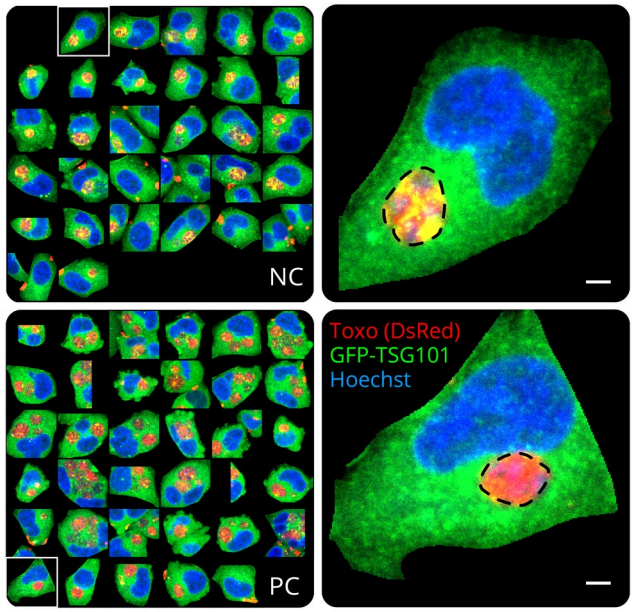



*Each cell is scored by the classifier; a well's score is the mean of its cells (preprint Fig. 2)*

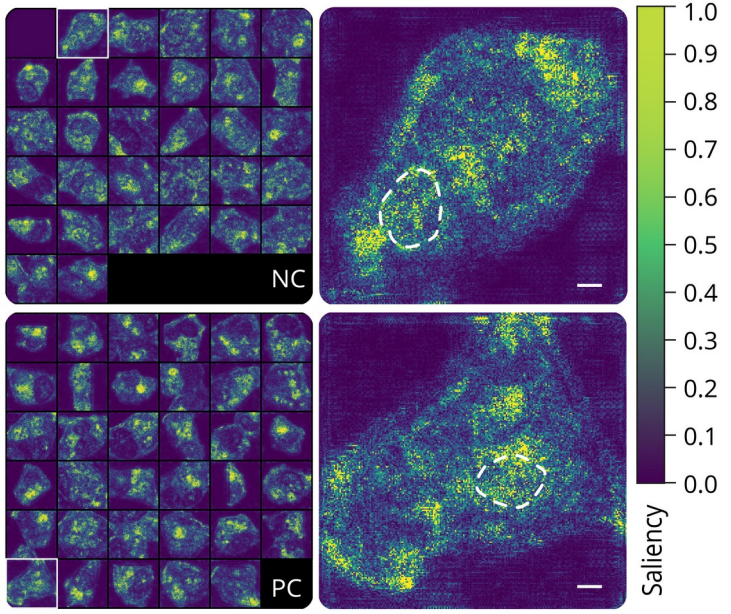
CORE · 4 OF 6

# Classify: image model

ALPHA




*Input: single-cell crops from negative (NC) and positive (PC) control wells*




*Saliency: where MaxViT looked: at the vacuole in PC cells*

## SUB-TOOLS INSIDE THIS MODULE

**Activation Maps**
Show which pixels drove each decision

**Classifier Evaluation**
Accuracy and mistakes per class and per plate, on unseen wells

**Training Runs**
Compare training curves between runs

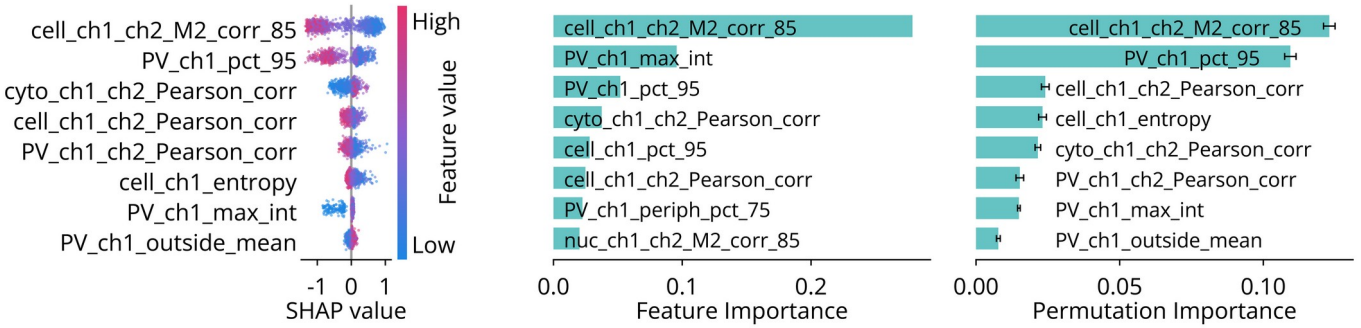
**Explain CV Model**
Explain an image model in terms of measurements

**Feature Explorer**
Which measurements separate the classes best

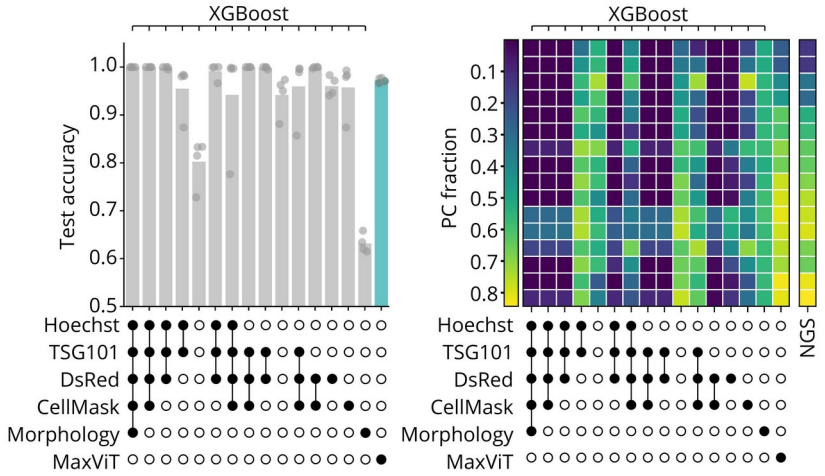
CORE · 4 OF 6

# Classify: measurement model

ALPHA




*XGBoost on the measurements: which measurements decide the class, three ways of ranking them*




*Accuracy with different channel combinations (grey, XGBoost) vs the image model (teal, MaxViT)*

CORE · 5 OF 6

# Map Barcodes: which guides, which well

STABLE

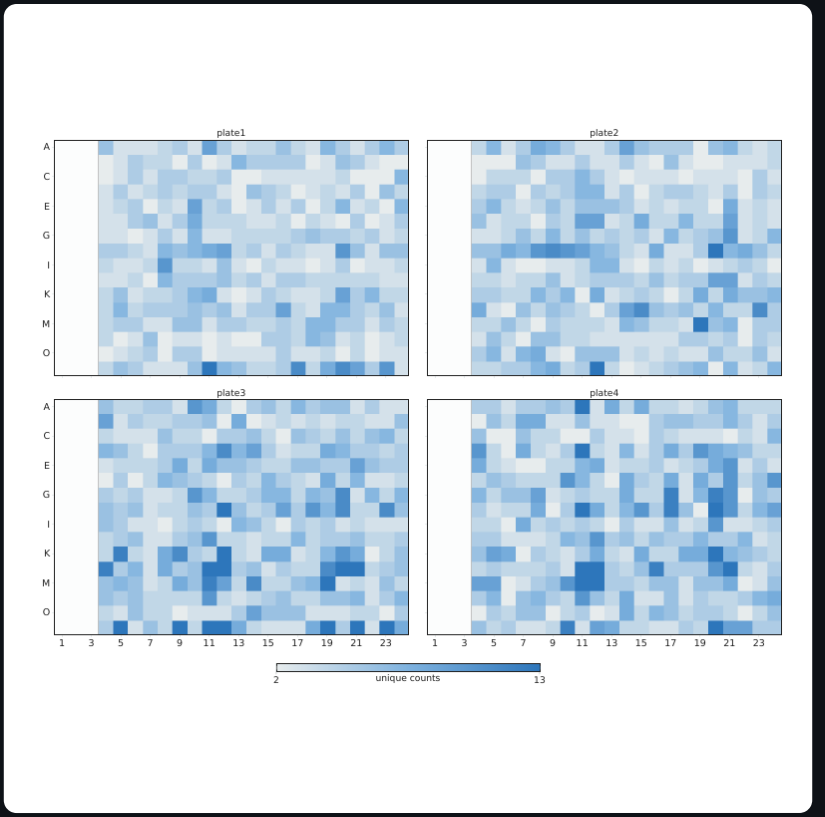
*Reads the sequencing data and works out which gRNAs are in each well, and how much of each.*

## WHAT YOU CAN DO

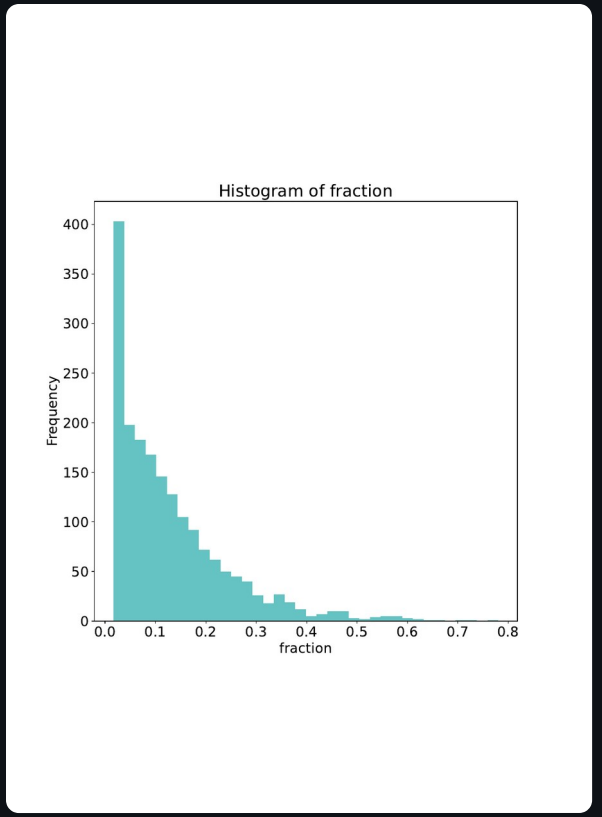
- Reads the row, column and gRNA barcodes from the sequencing files
- You give it the barcode lists and where they sit in the read
- Paired-end or single reads; set how many mismatches to allow
- Quick test on a small part of the file first

## OUTPUT

- Guide counts per well
- QC table and run manifest
- gRNA fraction per well → input to Regression




*Unique gRNAs per well, four plates*




*gRNA fraction per well*

## SUB-TOOLS INSIDE THIS MODULE

**Barcode QC**
Reads per well, reads that did not match, library coverage: and the cut-off that gives the expected number of gRNAs per well

CORE · 6 OF 6

# Regression: from wells to genes

STABLE

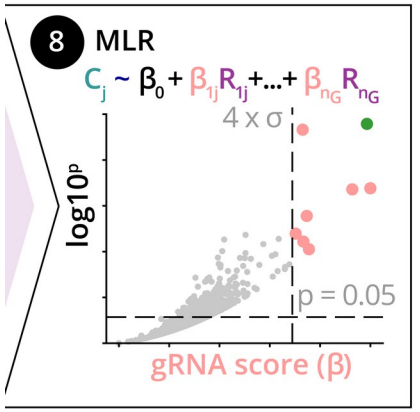
*Works out which genes change the phenotype, and ranks them.*

## WHAT YOU CAN DO

- Pairs each well's score with the fraction of each gRNA in that well
- Default: a mixed model where the gRNAs of a gene are grouped
- Other models for counts, fractions or skewed data
- Permutation test: shuffles gRNAs to check a hit is not chance

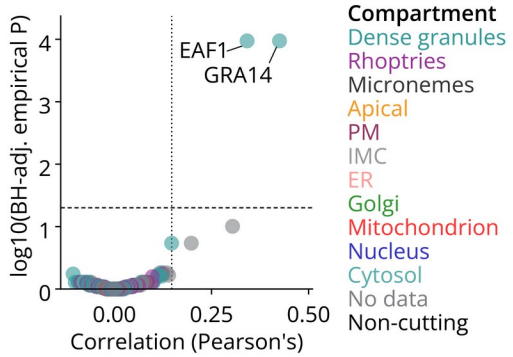
## OUTPUT

- A ranked table of genes with effect size and p-value
- Volcano, top effects and quality-check plots




G: gRNA
R: gRNA read *f*
C: Classifier score
β: gRNA score
$\beta_0$: Intercept
W: Well
i: gRNA index
j: Well index

*The model: a well's score = its gRNA fractions × each gene's effect (preprint Fig. 2)*




*The result: EAF1 and GRA14 stand out (preprint Fig. S7)*

CORE · 6 OF 6

# Regression: choose the model

STABLE

*21 model families, each explained in plain words, with what it assumes about your data.*

## WHAT YOU CAN DO

**Mixed:** the gRNAs of a gene grouped, genes allowed to vary

**OLS / WLS:** classic least squares, optionally weighted by cells per well

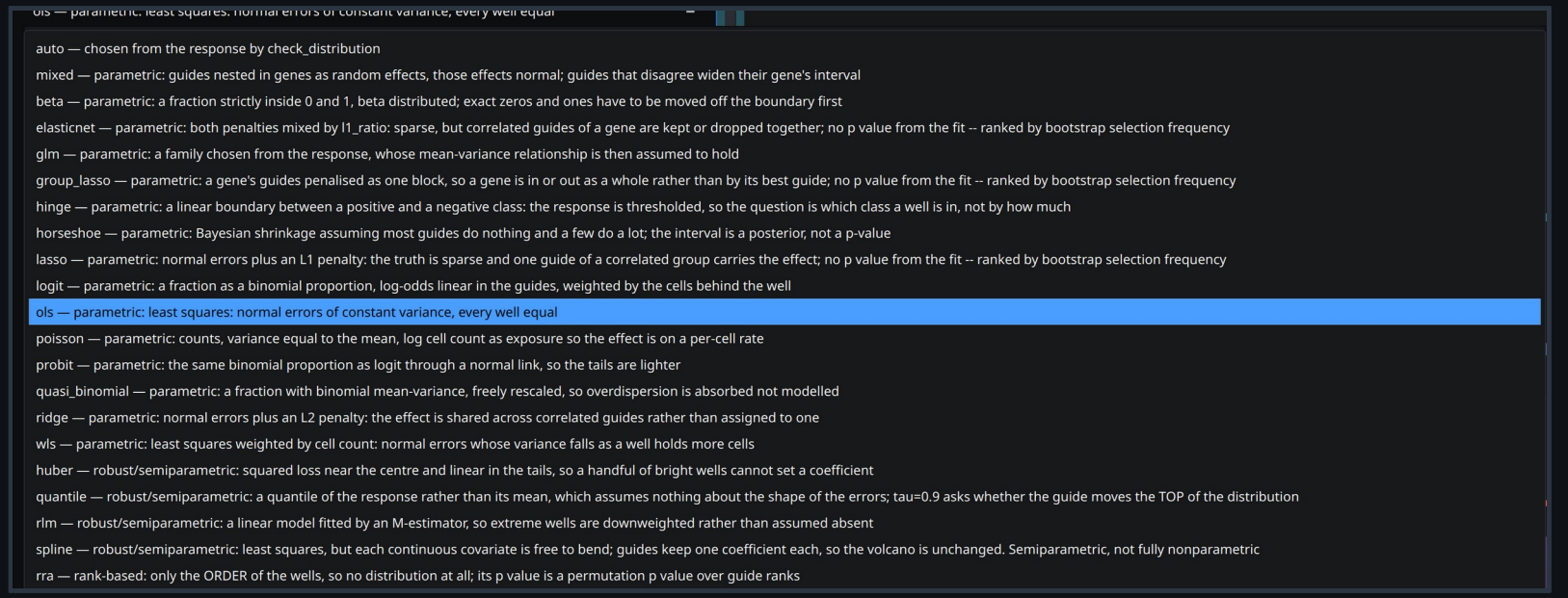
**GLM, Poisson, beta, logit:** for counts, fractions and yes/no readouts

**Lasso, ridge, elastic net, group lasso, horseshoe:** when there are many more genes than wells

**Robust, quantile, RRA:** when a few wells are extreme

## OUTPUT

- Plate, row and column covariates; permutation test on gRNAs
- The formula and what it assumes, shown for the chosen model

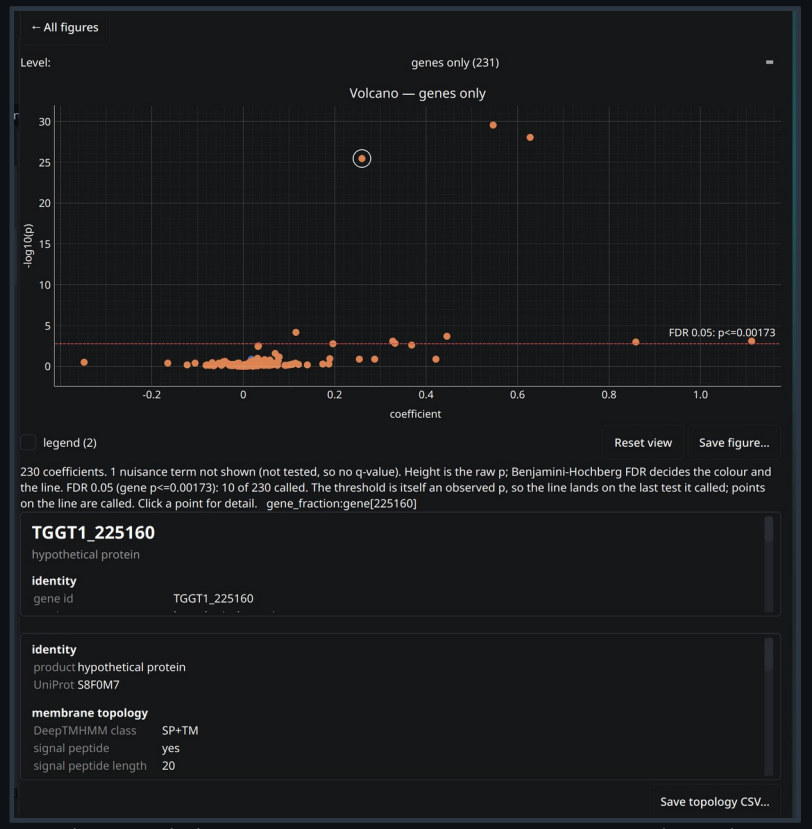



*Regression type: every model with the assumption it makes (OLS selected)*

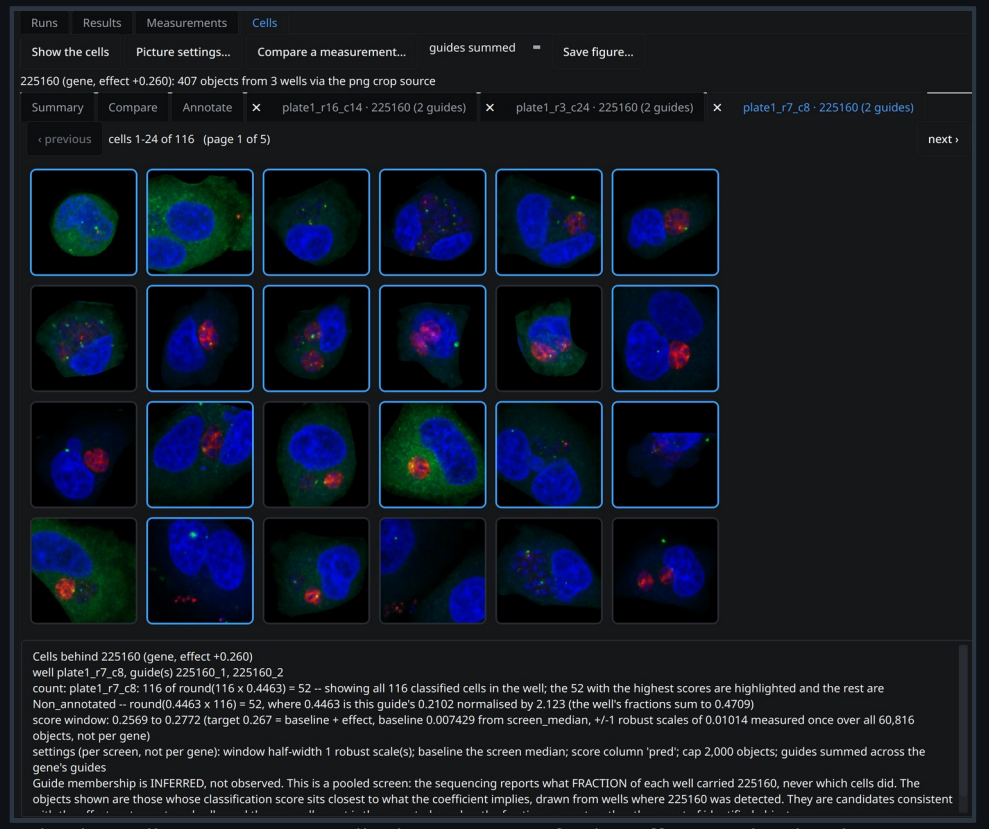
CORE · 6 OF 6

# Regression: pick a gene, see its cells

STABLE




*Interactive volcano: click a point. EAF1 (TGGT1_225160) is selected, its gene card below*




*Cells tab: the cells in an EAF1 well whose scores fit the effect, with why they were picked*

## SUB-TOOLS INSIDE THIS MODULE

**Volcano Explorer**
Publication version of the volcano; vector export

**Investigate Hit**
Test the hit at the cell level against control wells

**Hit List**
Ranked genes with intervals and gRNA agreement

**Prediction Profiler**
How the prediction changes as one gene's fraction changes

CORE · 6 OF 6

# Regression: which assumptions hold?

STABLE

*Every run checks its own assumptions and says, in red, which ones fail and what to do.*

## WHAT YOU CAN DO

**Equal variance:** Breusch-Pagan test on the residuals

**Normal residuals:** skew, kurtosis, D'Agostino test

**Independence:** Durbin-Watson; wells on one plate share handling

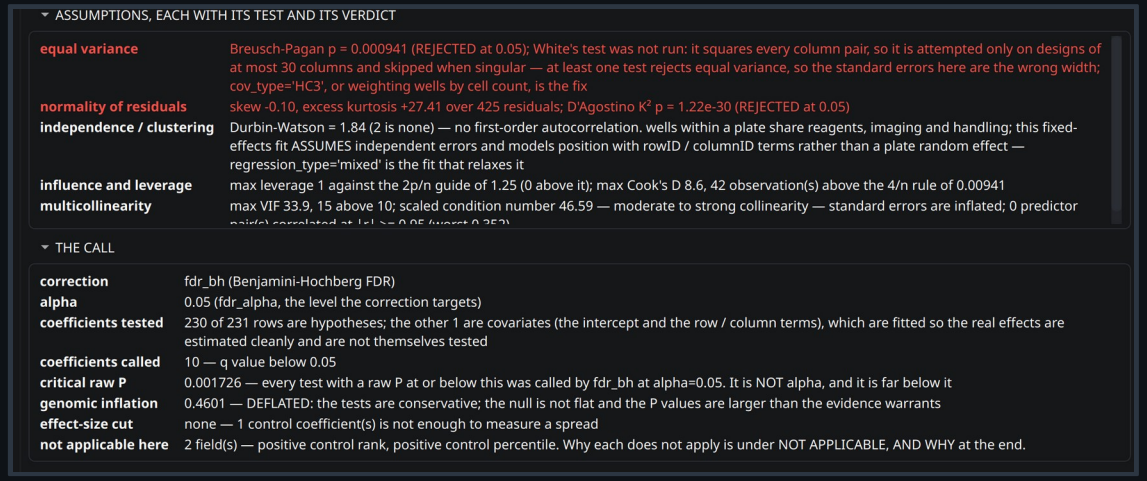
**Influence:** leverage and Cook's distance per well

**Collinearity:** VIF and condition number

**Calibration:** p-value histogram and genomic inflation

## OUTPUT

- A run summary: the answer, what was fitted, each assumption with its verdict
- Residual, Q-Q, scale-location and influence plots



ASSUMPTIONS, EACH WITH ITS TEST AND ITS VERDICT

| | |
|---|---|
| **equal variance** | Breusch-Pagan p = 0.000941 (REJECTED at 0.05); White's test was not run: it squares every column pair, so it is attempted only on designs of at most 30 columns and skipped when singular — at least one test rejects equal variance, so the standard errors here are the wrong width; cov_type='HC3', or weighting wells by cell count, is the fix |
| **normality of residuals** | skew -0.10, excess kurtosis +27.41 over 425 residuals; D'Agostino K² p = 1.22e-30 (REJECTED at 0.05) |
| **independence / clustering** | Durbin-Watson = 1.84 (2 is none) — no first-order autocorrelation. wells within a plate share reagents, imaging and handling; this fixed-effects fit ASSUMES independent errors and models position with rowID / columnID terms rather than a plate random effect — regression_type='mixed' is the fit that relaxes it |
| **influence and leverage** | max leverage 1 against the 2p/n guide of 1.25 (0 above it); max Cook's D 8.6, 42 observation(s) above the 4/n rule of 0.00941 |
| **multicollinearity** | max VIF 33.9, 15 above 10; scaled condition number 46.59 — moderate to strong collinearity — standard errors are inflated; 0 predictor pair(s) correlated at \|r\| >= 0.95 (worst 0.352) |

THE CALL

| | |
|---|---|
| **correction** | fdr_bh (Benjamini-Hochberg FDR) |
| **alpha** | 0.05 (fdr_alpha, the level the correction targets) |
| **coefficients tested** | 230 of 231 rows are hypotheses; the other 1 are covariates (the intercept and the row / column terms), which are fitted so the real effects are estimated cleanly and are not themselves tested |
| **coefficients called** | 10 — q value below 0.05 |
| **critical raw P** | 0.001726 — every test with a raw P at or below this was called by fdr_bh at alpha=0.05. It is NOT alpha, and it is far below it |
| **genomic inflation** | 0.4601 — DEFLATED: the tests are conservative; the null is not flat and the P values are larger than the evidence warrants |
| **effect-size cut** | none — 1 control coefficient(s) is not enough to measure a spread |
| **not applicable here** | 2 field(s) — positive control rank, positive control percentile. Why each does not apply is under NOT APPLICABLE, AND WHY at the end. |

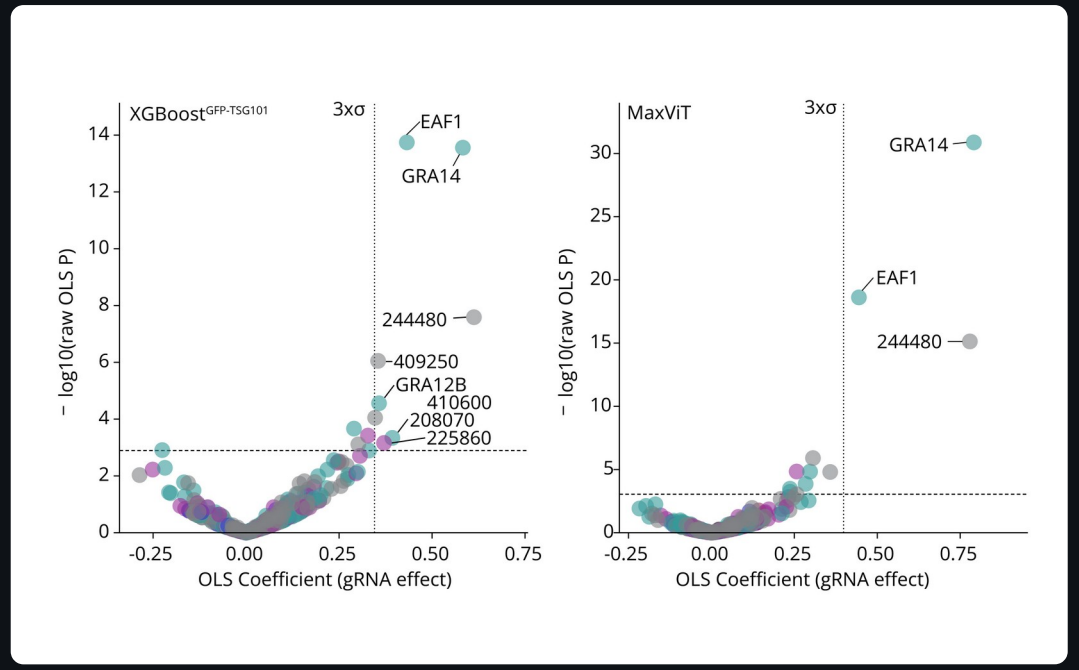
*Summary tab: each assumption with its test and verdict; equal variance and normality rejected (red)*

*The diagnostic plots behind the verdicts*

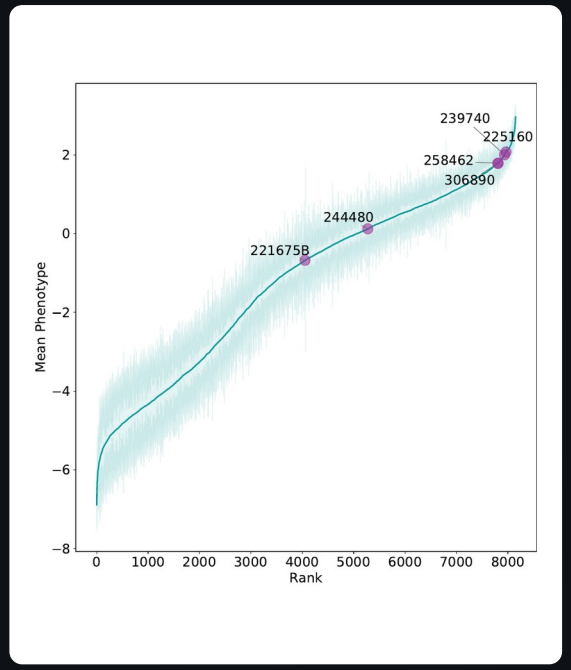
CORE · 6 OF 6

# Regression: the hit list

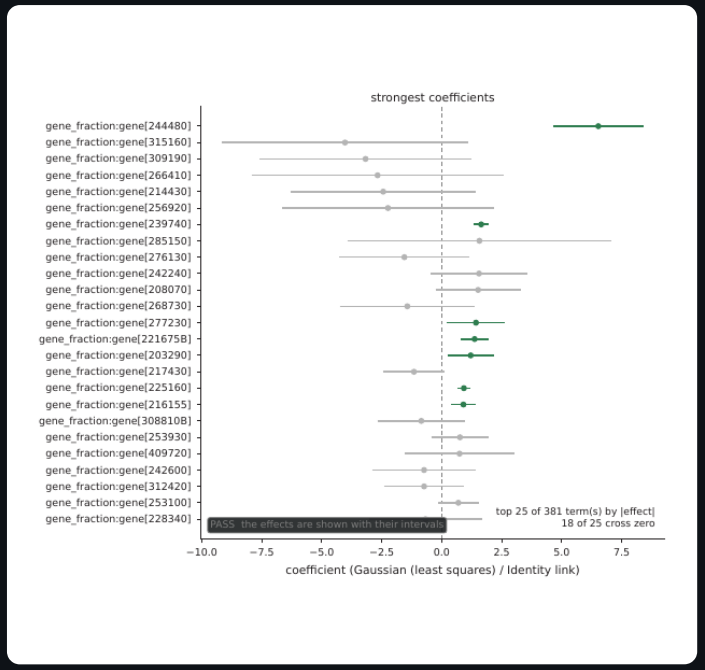
STABLE




*Volcanos from the screen: XGBoost (left) and MaxViT (right) scores both find EAF1 and GRA14 (preprint Fig. 5)*




*Genes ranked by phenotype*




*Strongest effects with 95% intervals*

## SUB-TOOLS INSIDE THIS MODULE

**Volcano Explorer**
Click a point to see the gene; export for publication

**Hit List**
Ranked genes with confidence intervals and gRNA agreement; export as a table

**Investigate Hit**
Jump from a gene to its wells and cells

**Diagnostics**
Checks that the model fits the data (plate, row and column effects)

**Methods & Results**
Drafts the methods text from the run itself

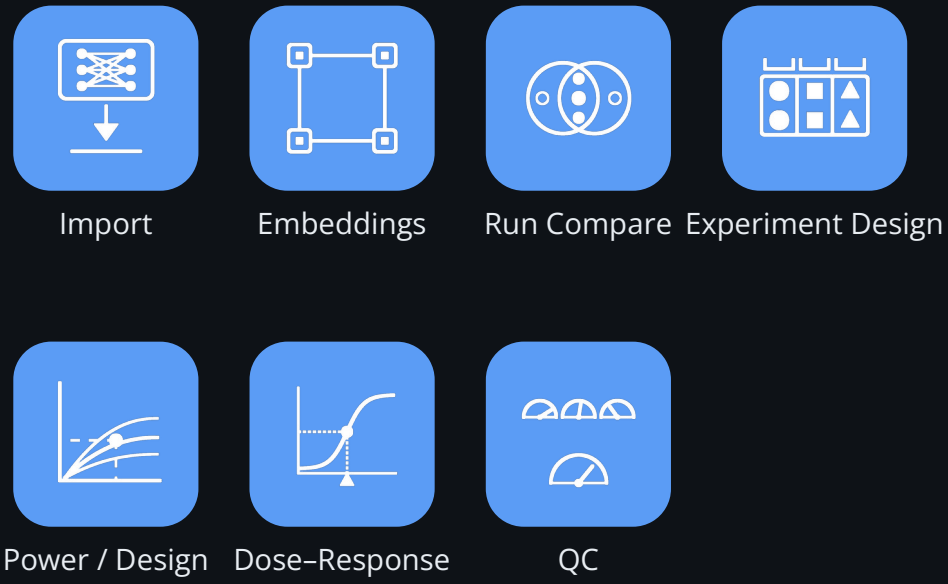
**Prediction Profiler**
How a prediction changes across one input

**Parameter Sweep**
Re-run across settings to see how stable the hits are

# 02

# Data

Bring data in, plan experiments, and check that a run can be trusted.



DATA · 1 OF 7

# Import: bring outside data in

STABLE

*Turns images, masks and tables from other tools into a spaCR project.*

## WHAT YOU CAN DO

- Pair images, instance masks and a measurement CSV
- Match their column names to spaCR's, converting units
- Choose what happens when names collide
- Preview before anything is written; save and reuse the mapping

## OUTPUT

- A spaCR project that records where the data came from

| Their column | spaCR column | Transform | Unit in | Unit out | Note |
|---|---|---|---|---|---|
| AreaShape_Area_um2 | foreign_areashape_area_um2 | area | um^2 | px^2 | unit um^2 read from the column name; spaCR measures in px^2, so this needs a pixel size |
| MeanIntensity_ER | foreign_meanintensity_er | identity | | | copied unchanged; declare unit_in/unit_out if it needs scaling |
| Metadata_Well | foreign_metadata_well | identity | | | copied unchanged; declare unit_in/unit_out if it needs scaling |
| cell_area | foreign_cell_area | identity | | | name suggests a area, but no unit is declared — set unit_in (and transform=area) if these are not already in px^2 |

*Column mapping: their name → spaCR column, with unit conversion*

| Source | Target | Plate | Well | Field | Channel | Z | T | From well | From field | From channel | Z handling | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| fov01_C1.tif | plate1_A01_T0001F001L01A01Z01C01.tif | plate1 | A01 | 1 | 1 | 1 | 1 | A01 | images/fov01 | C1 | keep | planned |
| fov01_C2.tif | plate1_A01_T0001F001L01A01Z01C02.tif | plate1 | A01 | 1 | 2 | 1 | 1 | A01 | images/fov01 | C2 | keep | planned |
| fov01_C3.tif | plate1_A01_T0001F001L01A01Z01C03.tif | plate1 | A01 | 1 | 3 | 1 | 1 | A01 | images/fov01 | C3 | keep | planned |
| fov01_C4.tif | plate1_A01_T0001F001L01A01Z01C04.tif | plate1 | A01 | 1 | 4 | 1 | 1 | A01 | images/fov01 | C4 | keep | planned |
| fov02_C1.tif | plate1_A01_T0001F002L01A01Z01C01.tif | plate1 | A01 | 2 | 1 | 1 | 1 | A01 | images/fov02 | C1 | keep | planned |
| fov02_C2.tif | plate1_A01_T0001F002L01A01Z01C02.tif | plate1 | A01 | 2 | 2 | 1 | 1 | A01 | images/fov02 | C2 | keep | planned |
| fov02_C3.tif | plate1_A01_T0001F002L01A01Z01C03.tif | plate1 | A01 | 2 | 3 | 1 | 1 | A01 | images/fov02 | C3 | keep | planned |
| fov02_C4.tif | plate1_A01_T0001F002L01A01Z01C04.tif | plate1 | A01 | 2 | 4 | 1 | 1 | A01 | images/fov02 | C4 | keep | planned |

*Format Converter: plan for turning vendor files into spaCR's TIFF layout*

## SUB-TOOLS INSIDE THIS MODULE

**Import Images**
Reads any microscope's folder and works out the naming itself

**Format Converter**
Turns Nikon, Zeiss and Leica files into spaCR's layout

**External Masks**
Measure masks made elsewhere, without re-segmenting

DATA · 2 OF 7

# Embeddings: a fingerprint per cell

ALPHA

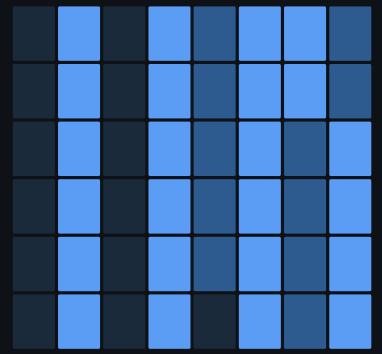
*Gives every cell a label-free feature vector from a pretrained image model, next to the measured features.*

## WHAT YOU CAN DO

- Crops from the project (filter by object class or plate) or from a folder
- Choose the network: ResNet-18 (default), ResNet-50, ConvNeXt, ViT
- Each stain is described separately, or all together
- Runs on the graphics card when there is one

*cell crops*

**pretrained backbone**
e.g. ResNet-18



*one row of numbers per cell*

## OUTPUT

- One row of numbers per cell, next to its measurements
- Which network was used is recorded

**Use it for** clustering and UMAPs without choosing features, finding cells that look like a hit, and checking whether a classifier is picking up something the measurements miss.

DATA · 3 OF 7

# Run Compare: what changed?

BETA

*Puts two runs of the same project side by side, so you know why the results moved.*

## WHAT YOU CAN DO

- Pick a project and any two of its runs
- Settings diff, grouped by settings panel
- Object, well and field counts per plate
- Hits that appeared, vanished or changed rank
- Incompatible runs are blocked unless you choose 'compare anyway'

## OUTPUT

- Side-by-side comparison tables

### Settings

| | |
|---|---|
| pathogen_model | cpsam → toxoplasma_pv_v3 |
| cell_diameter | 90 → 110 |
| regression_type | ols → mixed |

### Counts per plate

| | |
|---|---|
| cells | 182,410 → 176,902 |
| pathogens | 40,112 → 52,774 |
| wells | 384 → 384 |

### Hits

| | |
|---|---|
| appeared | 3 genes |
| vanished | 1 gene |
| rank changed | 12 genes |

*Illustration of the three comparison panels (values made up)*

DATA · 4 OF 7

# Experiment Design: plan the plate

ALPHA

*Lay out conditions, controls and replicates before you image anything.*

## WHAT YOU CAN DO

- Assign conditions and control roles to wells
- Warnings for capacity, too few replicates, and row / column / edge confounding
- Randomise with a seed, so the layout can be rebuilt exactly

## OUTPUT

- A plate map that later spaCR steps use directly
- The settings used, so the layout can be rebuilt

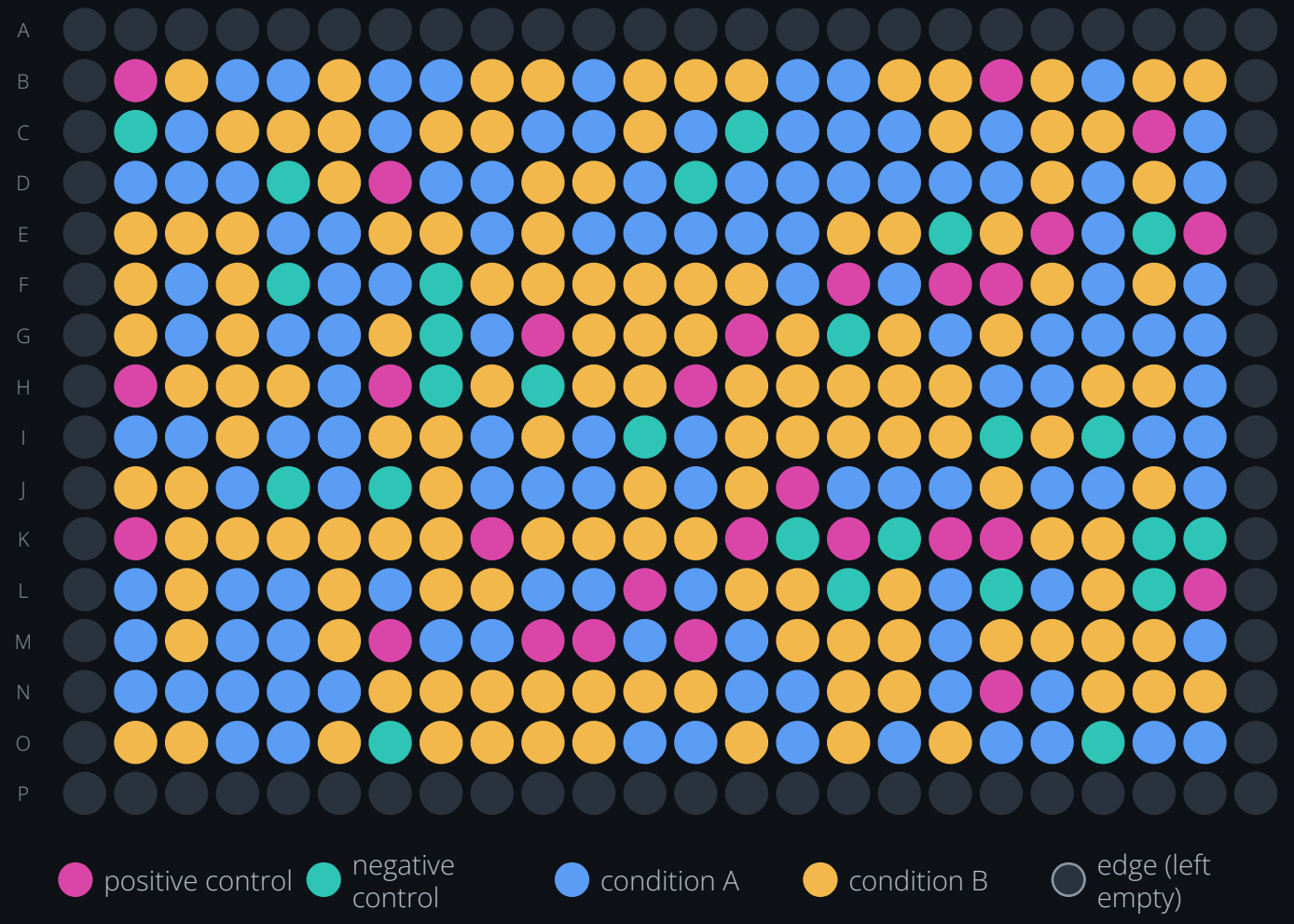



*Illustration: a randomised 384-well layout with edge wells left empty*

DATA · 5 OF 7

# Power / Design: how big a screen?

BETA

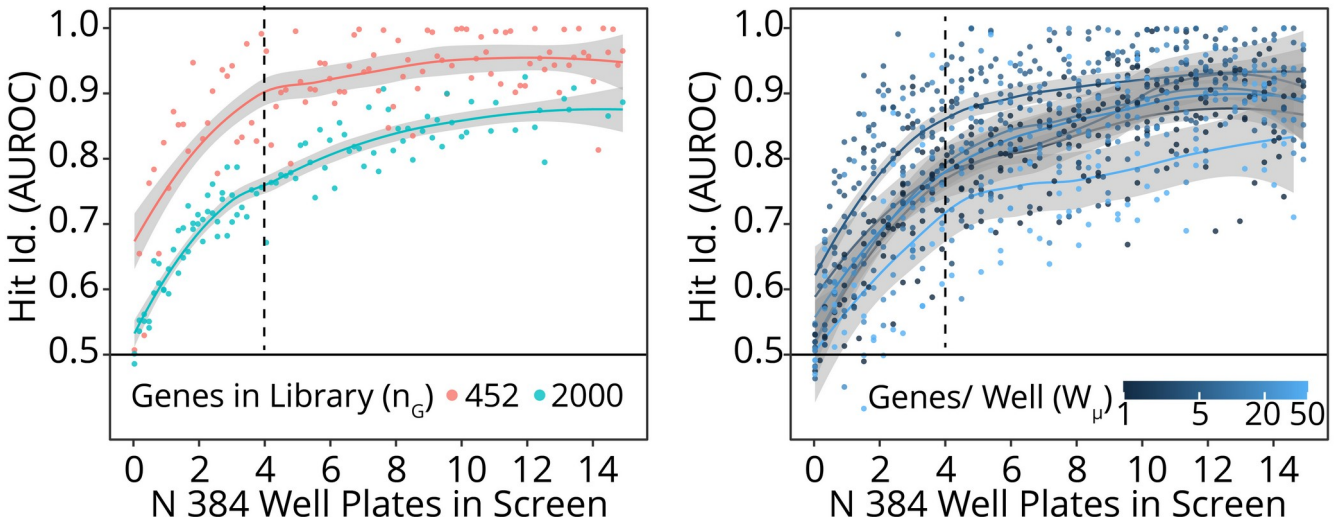
*Estimates how many cells per well and how many plates you need to find a real hit.*

## WHAT YOU CAN DO

- Enter the library: genes, gRNAs per gene, constructs per well
- Enter the plate format and number of plates
- Enter the classifier's background rate and the effect you expect
- Simulates the screen many times to see how often the real hit is found

## OUTPUT

- Detection probability vs cells per well
- A one-sentence answer, with caveats




*spaCRPower simulations: hit detection vs plates, by library size and genes per well (preprint Fig. 1D–E)*

DATA · 6 OF 7

# Dose–Response: honest EC50s

ALPHA

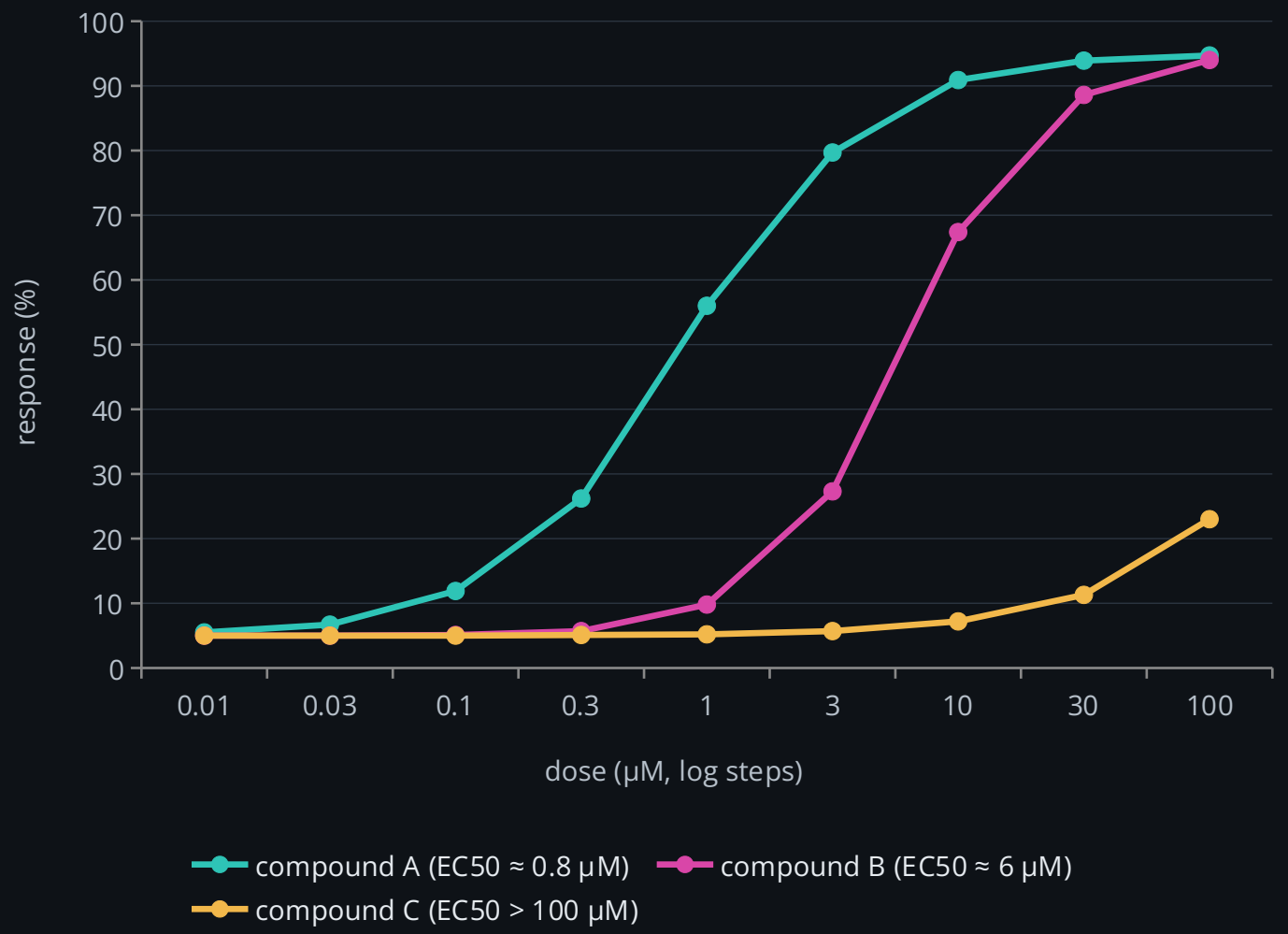
*Fits dose–response curves per gene or compound, and says so when an EC50 can't be pinned down.*

## WHAT YOU CAN DO

- Choose concentration, response and grouping columns
- Fits the standard S-shaped curve
- A confidence interval on every EC50

## OUTPUT

- Curves on a log-dose axis
- EC50 table with CI. Out-of-range → a bound ('EC50 > 30 µM'); failed fits listed as 'refused'
- Warns when a curve does not fit well




*Schematic of the output (simulated curves)*

DATA · 7 OF 7

# QC: is this run usable?

ALPHA

*Five stored checks on one page. Reads what the run already wrote; recomputes nothing.*

## WHAT YOU CAN DO

- Segmentation QC scorecards
- Pixel size recorded
- Classifier tested on truly unseen wells
- Plate effects (row / column / edge)
- Annotator agreement

## OUTPUT

- Verdict cards: ok · warn · fail · missing · stale
- Nothing is written to disk

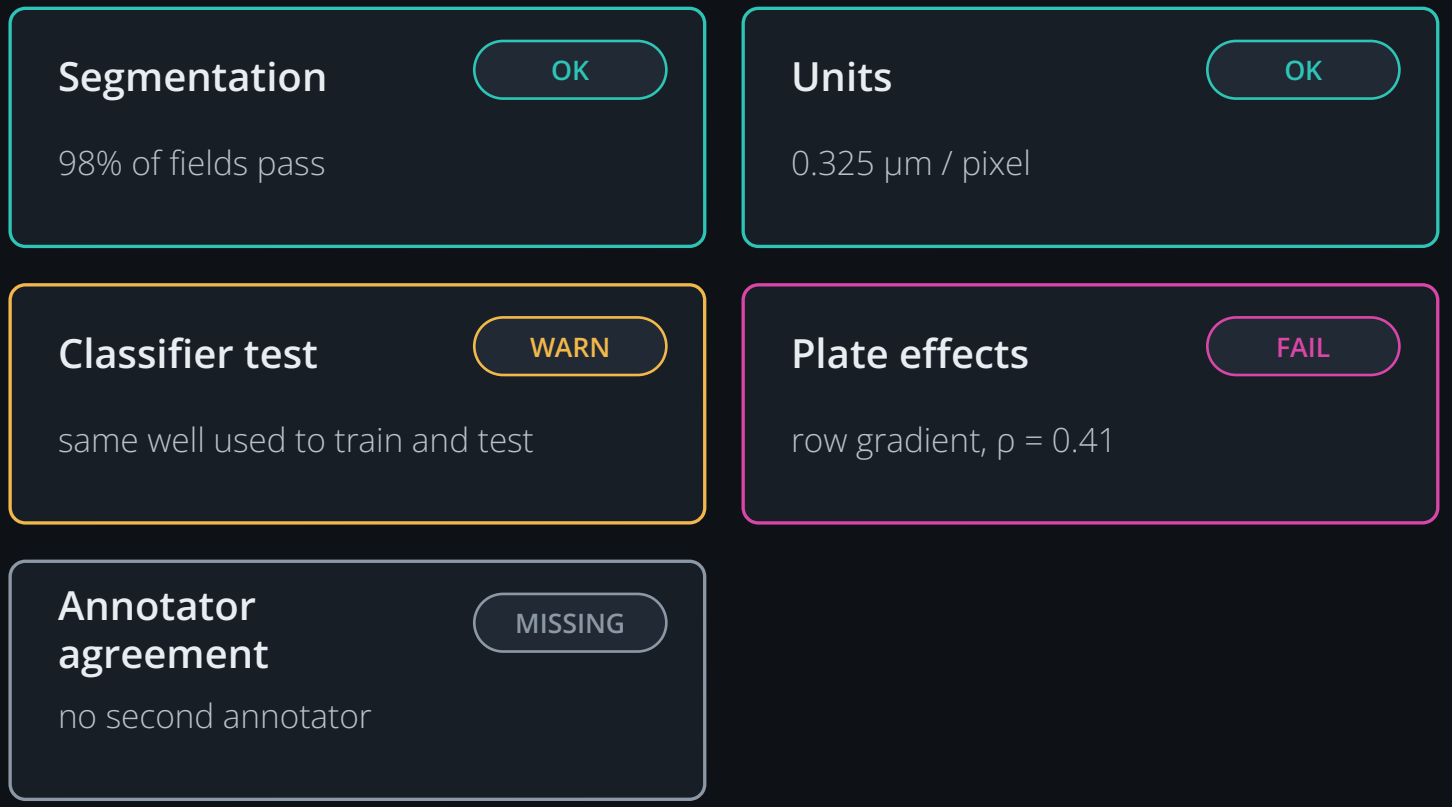



*Illustration of the dashboard's verdict cards (values made up)*

## SUB-TOOLS INSIDE THIS MODULE

**Layer Viewer**
Images and masks as layers you can switch on and off

**Control Charts**
Follow a control over time and flag drift

**Outliers**
Find unusual cells or wells

# 03

# Tools

Point these at a project: fix masks, stitch tiles, explore cells, gate populations, plot.

Make Masks

Align & Stitch

Image UMAP

Gate Editor

Graph Builder

TOOLS · 1 OF 5

# Make Masks: edit labels and boxes

BETA

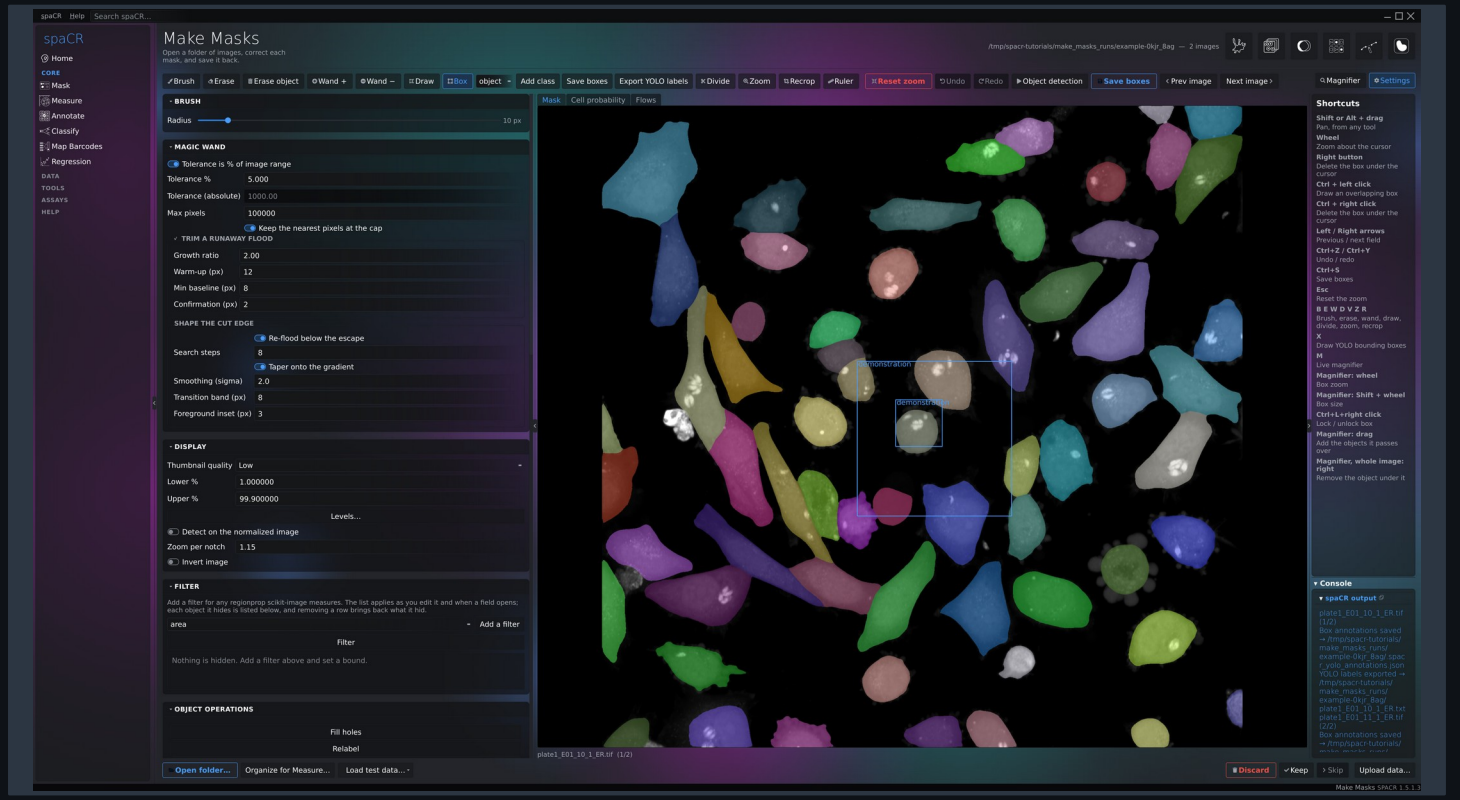
*Correct instance masks, prepare training pairs, or annotate independent YOLO boxes.*

## WHAT YOU CAN DO

- Mask tools: brush, erase, magic wand, draw, divide, zoom, recrop and ruler
- TEM images (DM3, DM4, MRC): an Electron microscopy contrast preset and TEM test data
- Detect Cellpose masks, threshold objects or puncta inside parent masks
- Add classes; draw, move and resize boxes with Undo and Redo
- Open a folder, Organize for Measure or Load test data; Discard, Keep, Skip or Upload data per field

## OUTPUT

- Curated masks, ready for training or measuring
- Saved box projects and normalized YOLO labels; source images stay unchanged

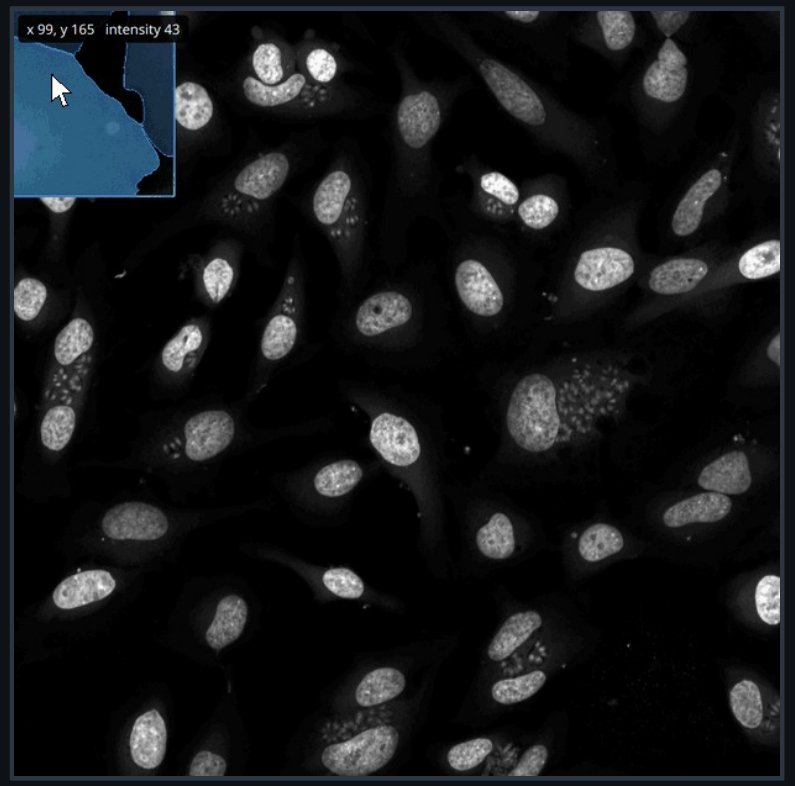



*Native Make Masks: demonstration boxes on an acquired microscopy field*

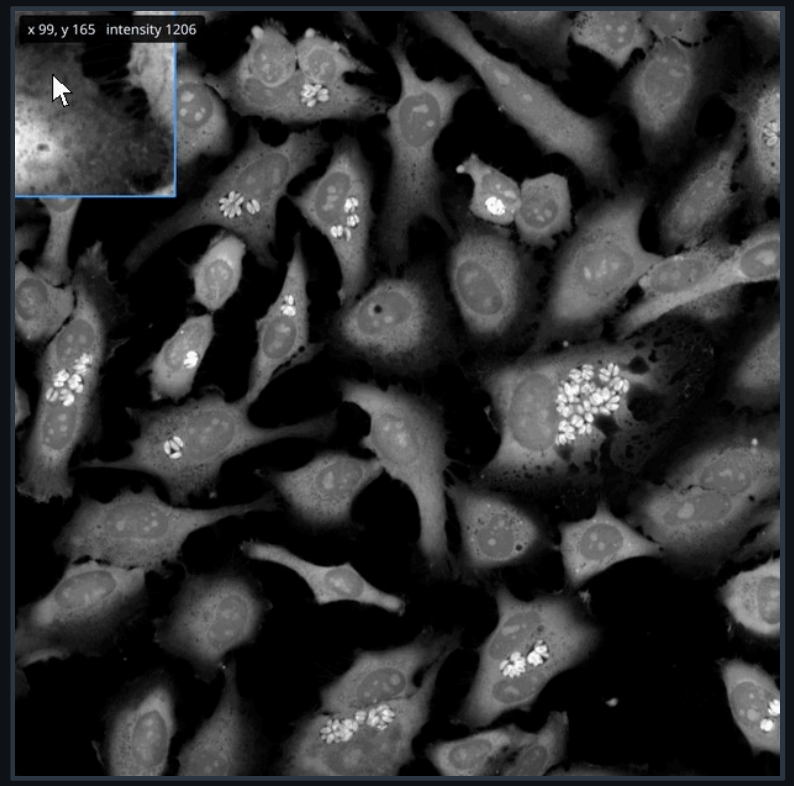
TOOLS · 1 OF 5

# Make Masks: live magnifier

BETA




*Cell from Hoechst: whole cells found from the nuclear (Hoechst) image*




*Toxoplasma from Cell Mask: vacuoles found from the cell-mask image*

**Box**
follows the mouse, zoomed and contrast-adjusted

**Model**
Cellpose-SAM, your own model, or a threshold, outlining live

**Click**
adds what the box found; every click can be undone

**Drag**
along a cell bigger than the box joins its pieces into one object

TOOLS · 1 OF 5

# Make Masks: organize for Measure

BETA

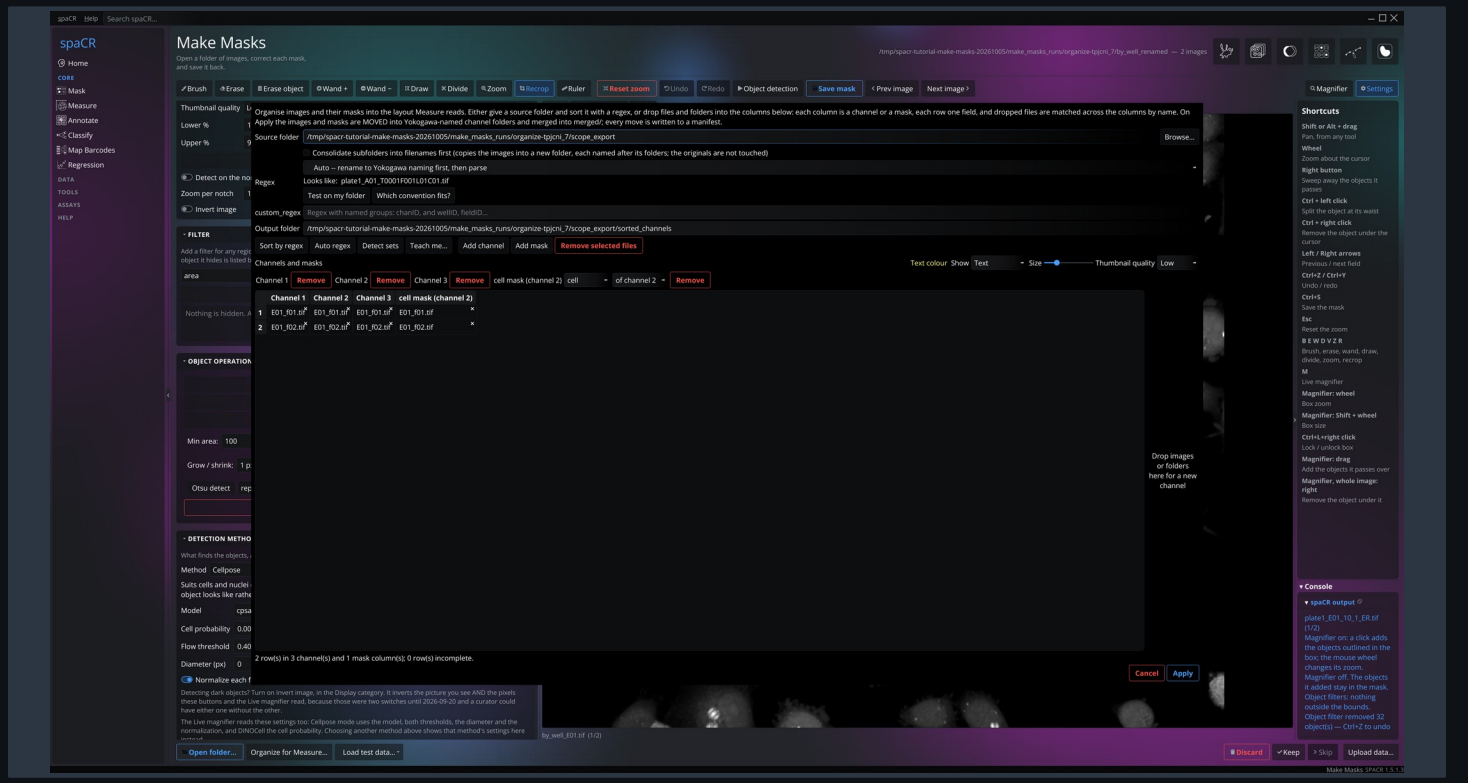
*Prepare images and corrected masks for the Measure module.*

## WHAT YOU CAN DO

- One row per field; columns are the image channels and the mask types
- Drop files, select files in the table, or detect sets from their names
- Apply the layout, then open Measure using the prepared merged arrays

## OUTPUT

- Channel-sorted images, merged arrays and an auditable move manifest

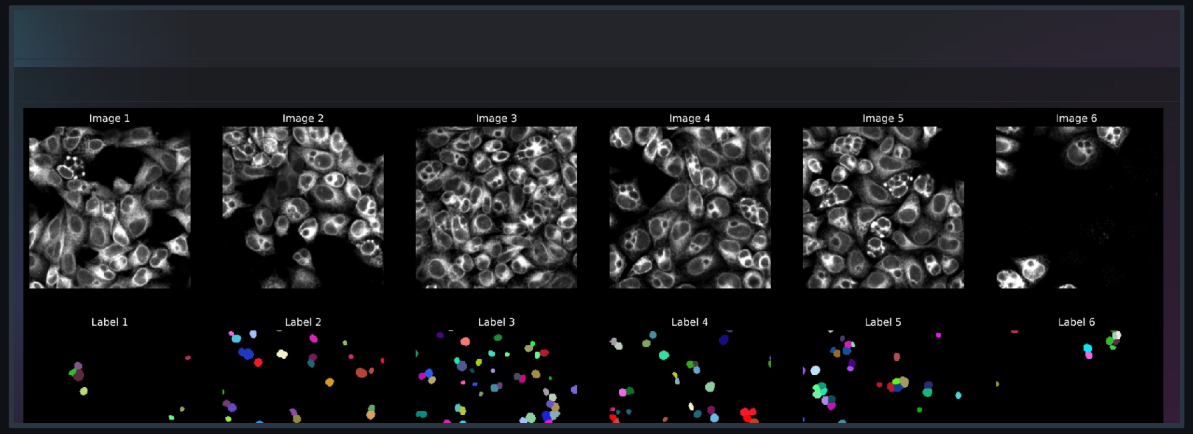



*Two acquired fields with three image channels and their cell masks*

TOOLS · 1 OF 5

# Make Masks: train and compare models

BETA




*Cellpose Workbench, training: image / label pairs it will learn from*

*Cellpose Workbench, applying a model: image, mask overlay and flows*

## SUB-TOOLS INSIDE THIS MODULE

**Cellpose Workbench**
Fine-tune Cellpose on your masks, then mask the whole folder

**Model Compare**
Two models' masks side by side, and how much they agree

**Model Zoo**
Browse, download and test trained models

**Curate**
Edit masks and tracks, with an edit log

**Napari Bridge**
Edit a mask in napari and bring it back

TOOLS · 2 OF 5

# Align & Stitch: tiles to mosaic

STABLE

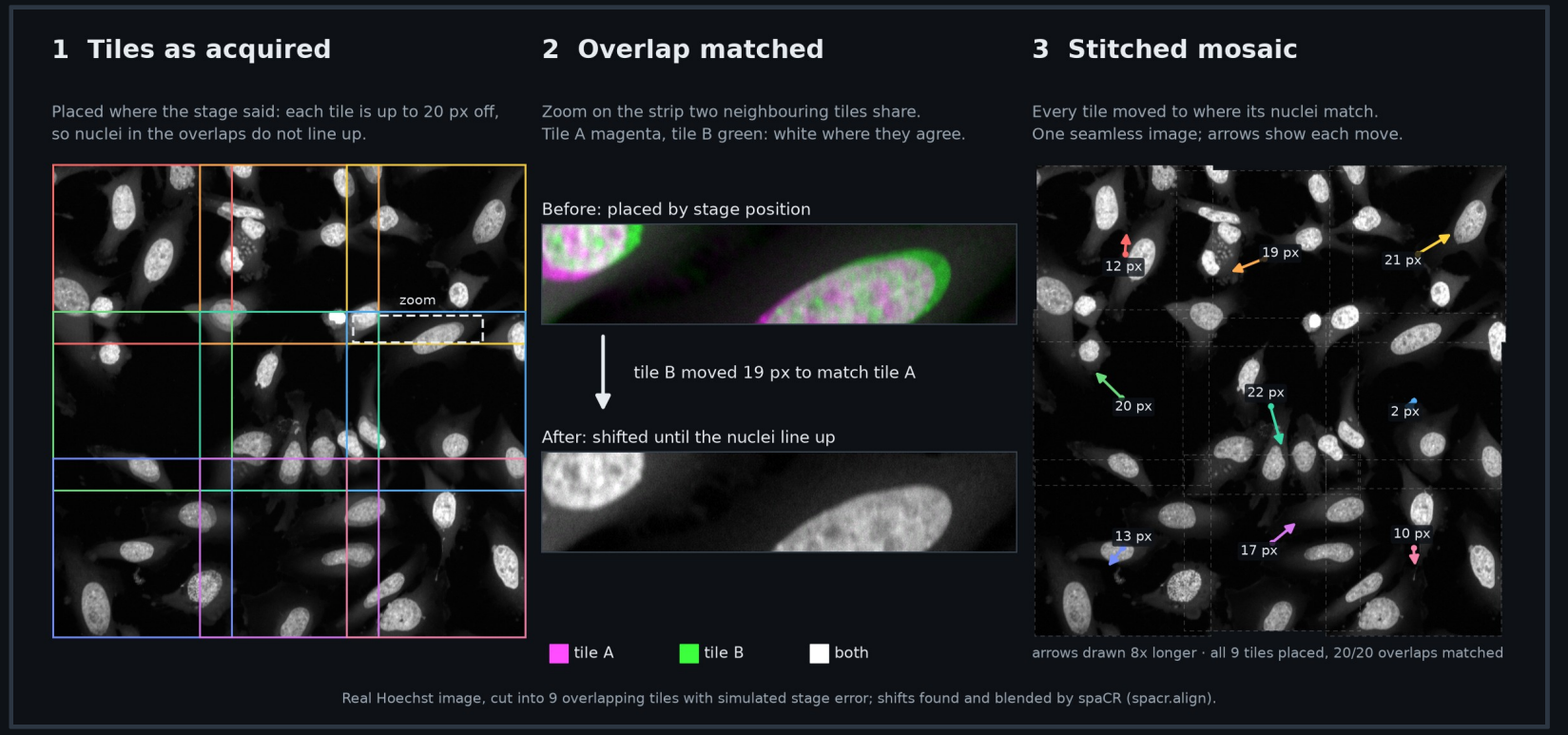
*Stitches overlapping tiles into one large image without running out of memory.*

## WHAT YOU CAN DO

- Set grid, overlap, tile order and reference channel
- Plans first, reading only the overlaps: fast on huge plates
- Finds how far each tile is off by matching nuclei in the overlaps
- Shows how sure it is about every tile
- Tells you the final size and memory needed before it starts

## OUTPUT

- The stitched mosaic, written piece by piece so memory stays low
- Where every tile ended up, saved with the project




*A real Hoechst image cut into 9 overlapping tiles, each up to 20 px off (simulated); spaCR put every tile back where it came from*

TOOLS · 3 OF 5

# Image UMAP: map all cells

BETA

*Places every cell on a 2-D map where similar cells sit together: with the real images.*

## WHAT YOU CAN DO

- UMAP, t-SNE or PCA maps
- Finds groups of similar cells automatically
- Embed by controls (positive vs negative groups)
- Can choose the map settings for you
- Click a point to see that cell's crop

## OUTPUT

- Scatter coloured by cluster or condition, with thumbnails
- Table with a cluster column per cell

*5,000 real cells from TSG101 screen plate 1: they fall into groups by number of nuclei and vacuoles*

## SUB-TOOLS INSIDE THIS MODULE

**Image Scatter**
Any two measured features, with the crop shown on hover

**PCA**
Which measurements drive the main differences

TOOLS · 4 OF 5

# Gate Editor: gate cells like FACS

ALPHA

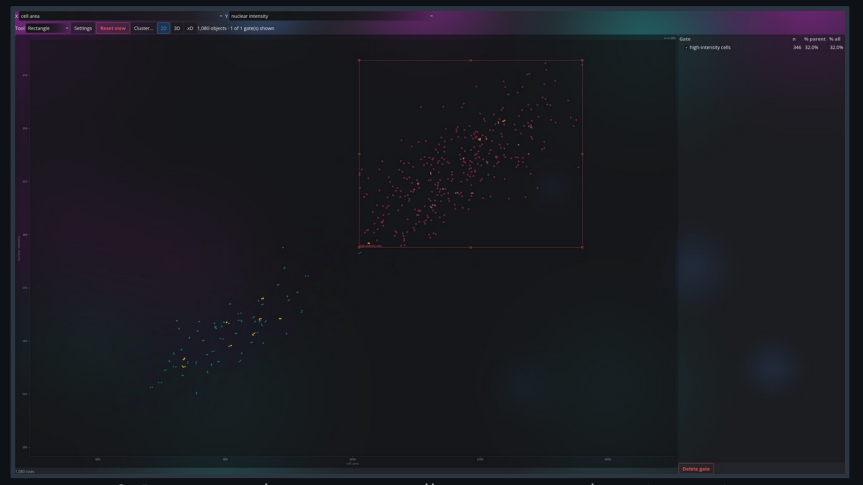
*Select cell populations the way a flow cytometrist gates: in 1, 2 or 3 dimensions.*

## WHAT YOU CAN DO

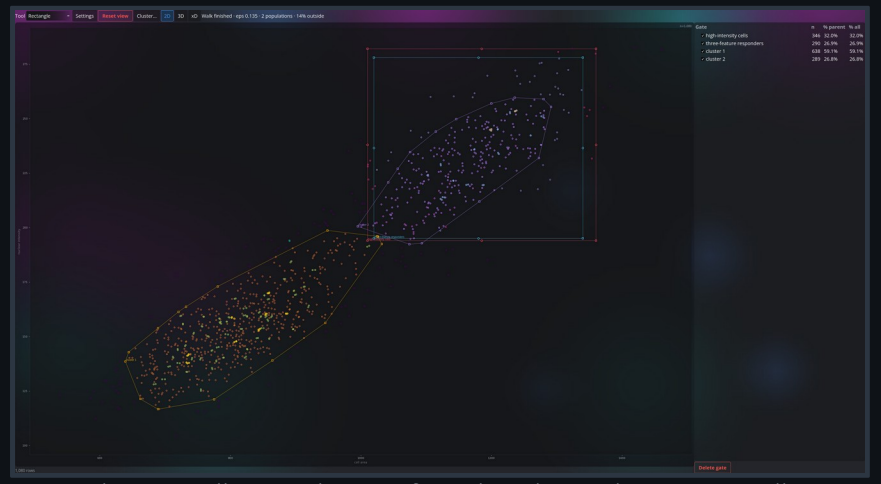
- Load tables from one or several projects, or a CSV
- Threshold, rectangle, oval, polygon, wand gates in 1-D and 2-D
- Box, cylinder and prism gates in 3-D
- Nest and combine gates; count and % of parent for each
- Publish a gate as a filter every linked view follows

## OUTPUT

- A gating strategy you can reuse on other plates
- Gate-membership columns in the project; PNG / PDF plots



*2-D rectangle gate on cell area vs nuclear intensity*



*Cluster walk: populations found and gated automatically*

TOOLS · 5 OF 5

# Graph Builder: plots without code

BETA

*Build a plot by dragging columns onto axes. The chart type follows the data.*

## WHAT YOU CAN DO

- Six drop zones: x, y, colour, size and two facets
- Histogram, bar, scatter, box or heatmap chosen for you; violin and line on request
- Brush points to select cells: linked to the UMAP and plate map
- Annotate conditions from metadata; reusable schemas
- Merge tables by name; save a chart and load it back

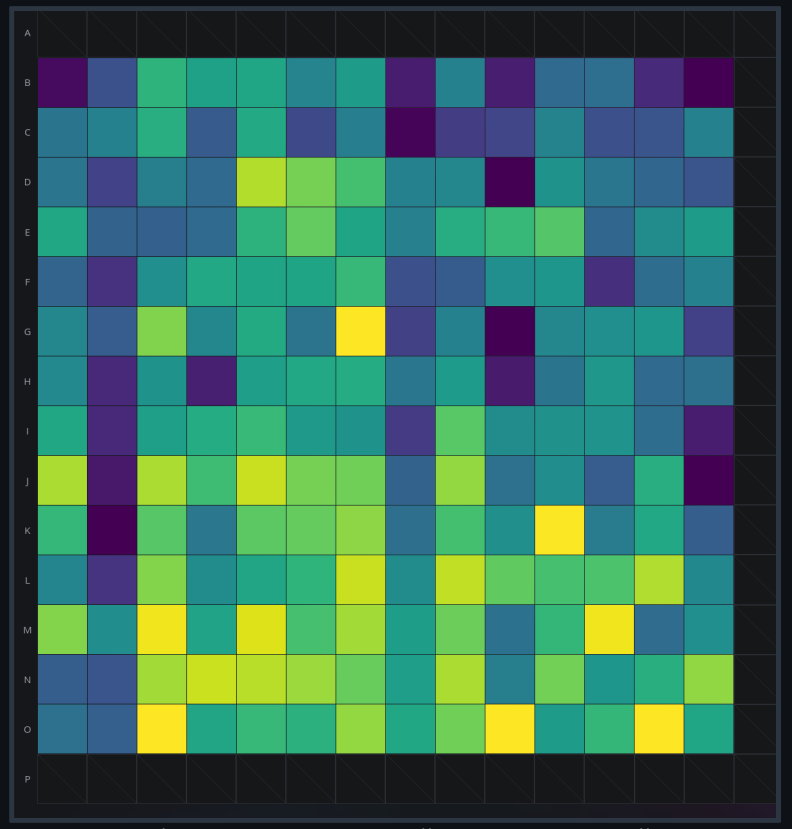
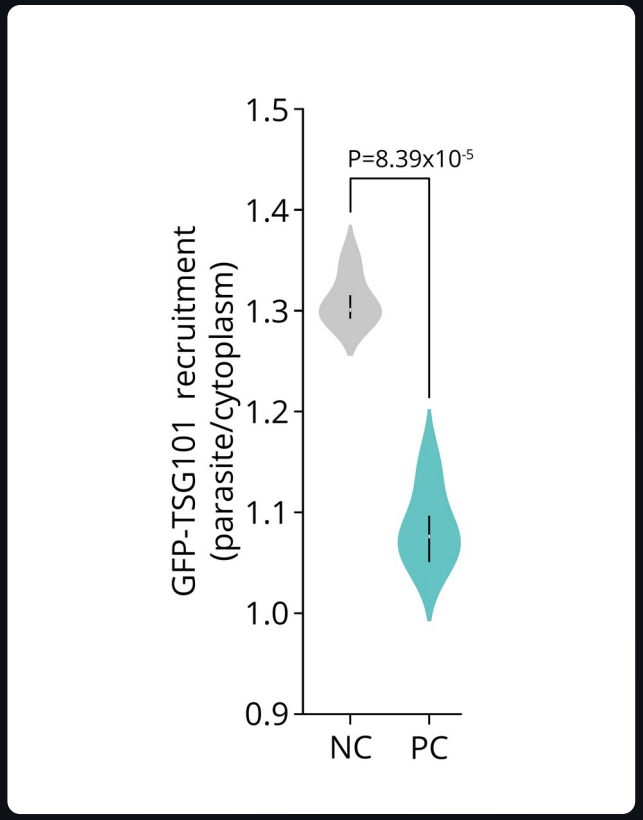


*Plate Viewer: mean cell intensity per well*




*Violin plot of per-well values*

## OUTPUT

- Interactive charts and annotated tables, exportable

## SUB-TOOLS INSIDE THIS MODULE

**Plate Viewer**
Plate heatmaps and edge-effect detection

**Small Multiples**
One plot per group, shared or independent axes

04

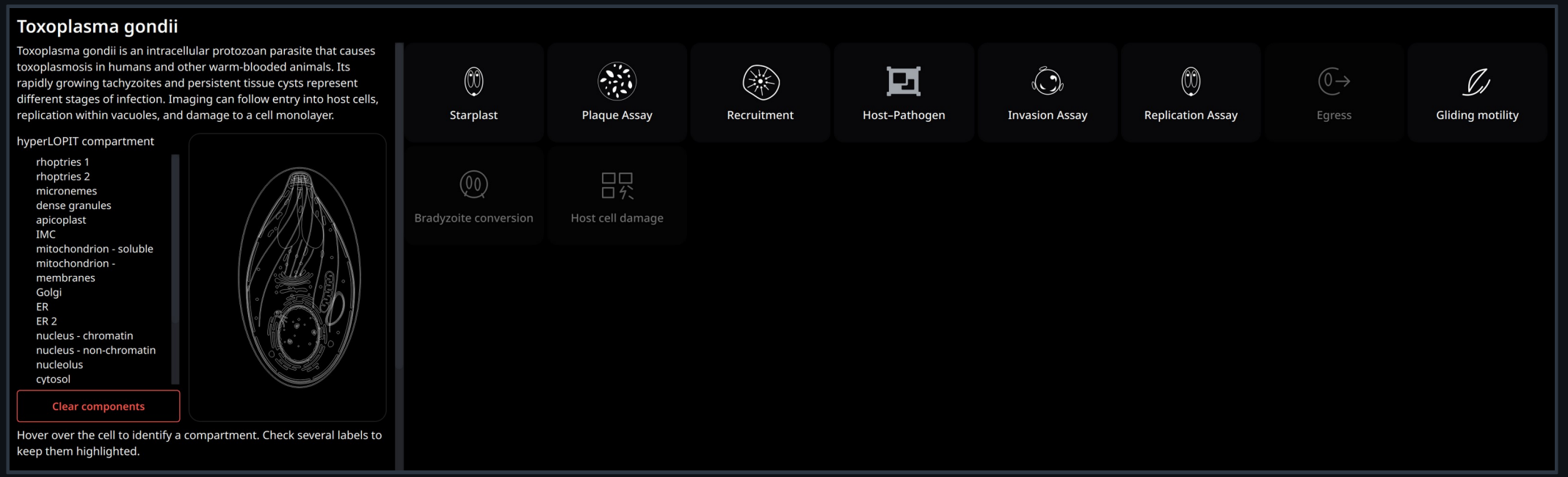
# Assays

One module per organism, each with its own assays. Toxoplasma ships today; more organisms are alpha species.

Toxoplasma

ASSAYS

# One module per organism




*The Toxoplasma module: the parasite's compartments on the left, its assays as tiles*

## Toxoplasma

Plaque · Recruitment · Host–Pathogen · Invasion · Replication · Gliding motility, with Starplast's hyperLOPIT compartments

## More organisms (alpha)

Plasmodium, Candida, Trypanosoma, Leishmania, Giardia, virus and mammalian modules: Preferences > Show alpha species

*Greyed tiles are assays still being built.*

ASSAYS · TOXOPLASMA · 1 OF 4

# Plaque Assay: count and size plaques

BETA

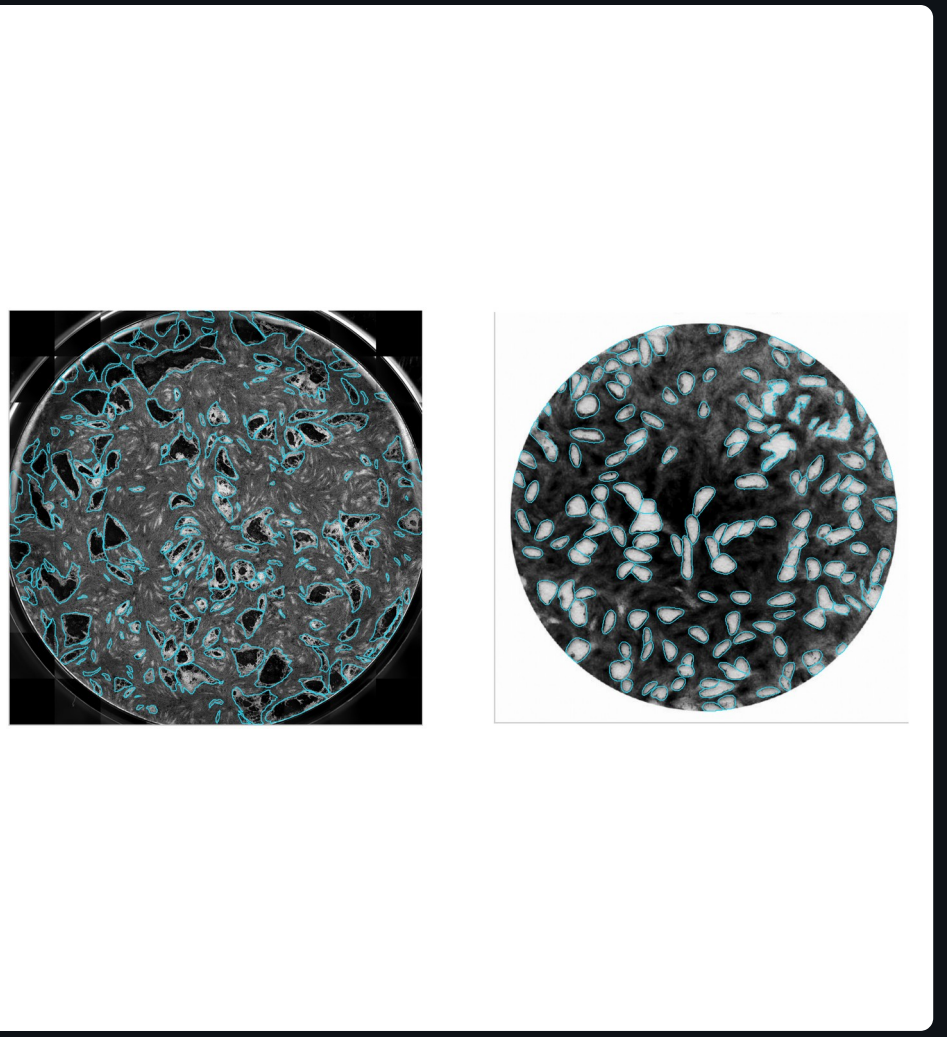
*Finds every plaque in whole-well images and measures it in mm².*

## WHAT YOU CAN DO

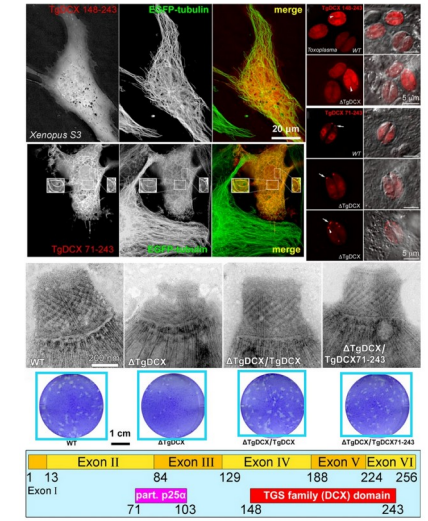
- Segments plaques with the Model Zoo plaque model (Cellpose-SAM)
- A YOLO detector finds and splits wells first
- Areas converted to mm² from the well diameter
- 'Figure' mode for plaque images taken from papers
- Can reuse existing masks

## OUTPUT

- A table of every plaque, and a summary per well
- Area, perimeter, diameter, eccentricity, solidity per plaque



*Curated plaque outlines: our microscope (left) and a published figure (right)*




*Wells found in a figure*

ASSAYS · TOXOPLASMA · 2 OF 4

# Recruitment: marker at the vacuole

STABLE

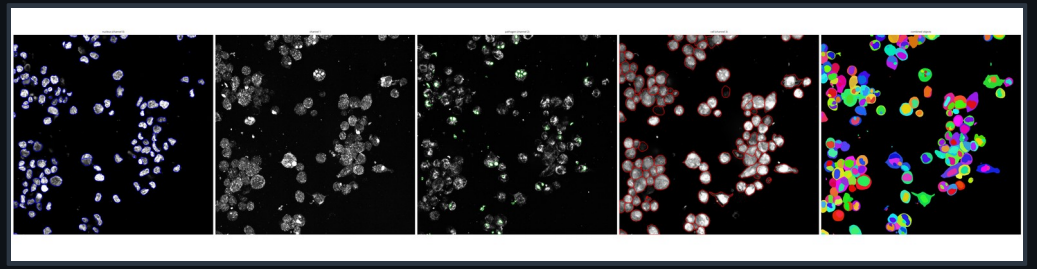
*How much does a fluorescent marker gather at the parasite vacuole compared with the host cytoplasm?*

## WHAT YOU CAN DO

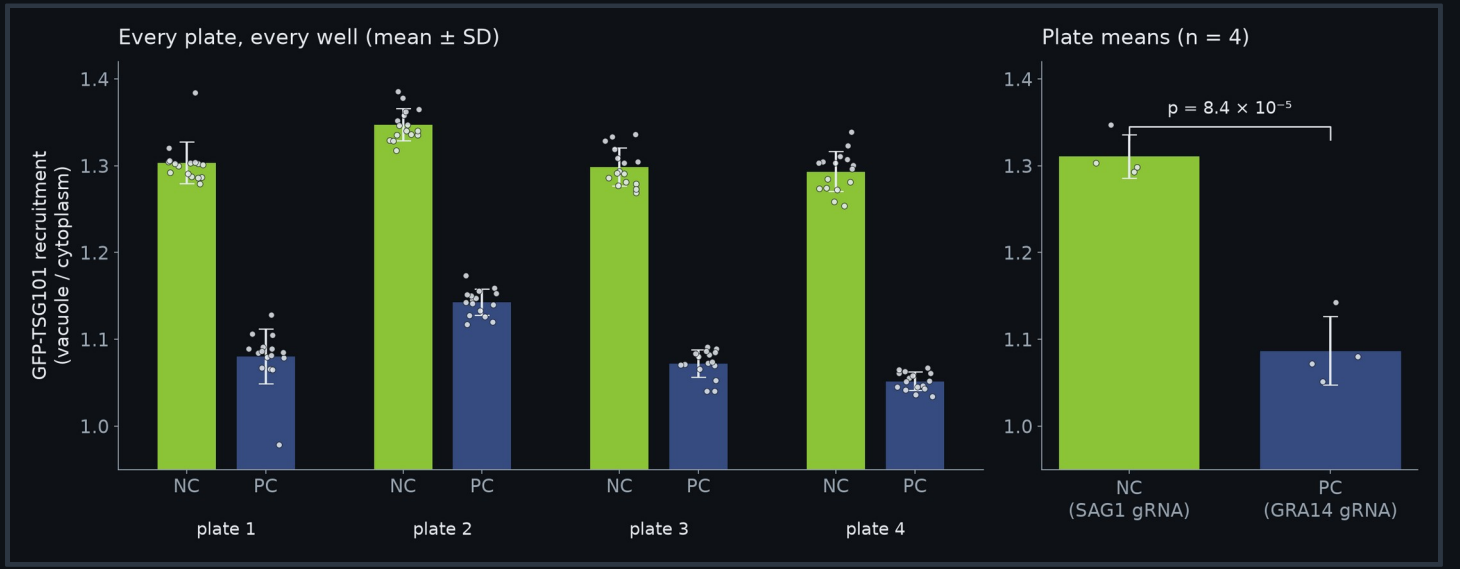
- Map wells to cell type, pathogen and treatment
- Size and intensity filters per object class
- Choose the channel of interest
- Minimum number of cells per well

## OUTPUT

- Tables per cell and per well
- Bar plots of recruitment by condition, with per-well points



*One field: channels and the cell / nucleus / parasite masks*




*GFP-TSG101 recruitment to the vacuole, NC vs PC, 4 plates (preprint data)*

ASSAYS · TOXOPLASMA · 3 OF 4

# Invasion: attached or invaded?

STABLE

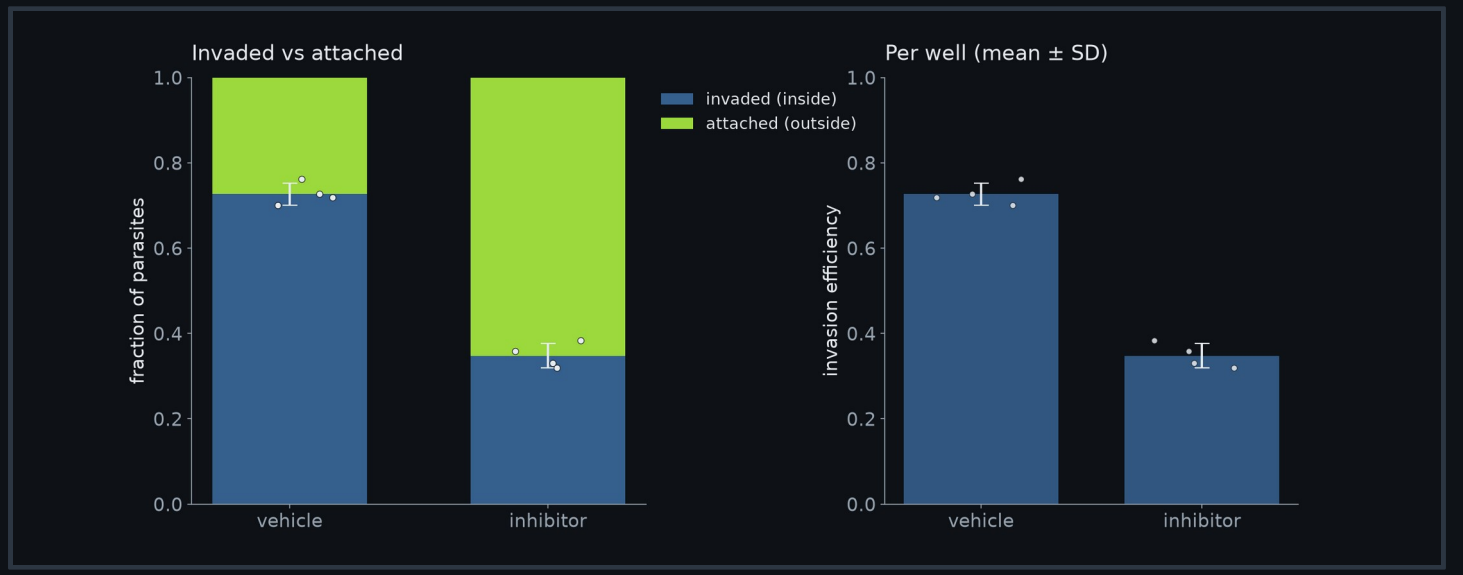
*Scores each parasite as outside (attached) or inside (invaded) from two-colour differential staining.*

## WHAT YOU CAN DO

- Set the outside-stain and total-stain channels
- Thresholds per field, or from control wells
- Flags fields where inside and outside cannot be told apart
- Checks the answer holds if the threshold moves

## OUTPUT

- Per-parasite calls, per-field thresholds, per-well invasion
- Statistics comparing conditions, per well
- Plots per well, by condition, and of the thresholds




*spaCR invasion analysis of the test dataset (4 wells per condition, mean ± SD)*

# Replication: parasites per vacuole

BETA

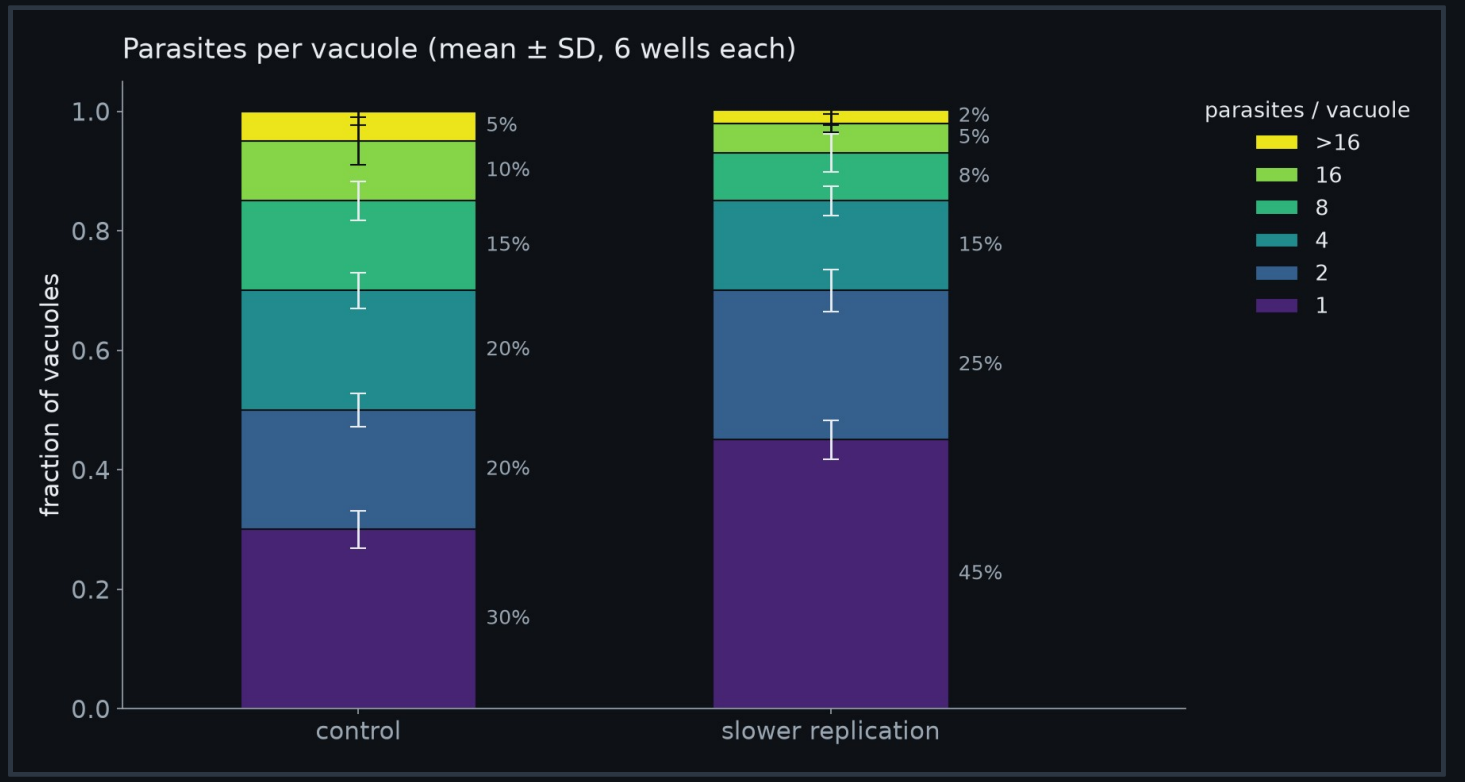
*Counts parasites per vacuole (1, 2, 4, 8, 16 …) and compares the distributions between conditions.*

## WHAT YOU CAN DO

- Groups parasites into vacuoles: auto, spatial, by cell, or by object
- Area filters for parasites and vacuoles
- Warns when too many vacuoles have a non-power-of-two count

## OUTPUT

- Vacuole counts, per-well distributions, condition summary
- Statistics comparing conditions
- Stacked-bar plots per well and by condition




*Example output on synthetic data: 6 wells per condition, mean ± SD*

**In progress:** a classification model that finds and counts Toxoplasma from the Toxoplasma stain, from the nuclear stain, or from the cell mask, so parasites per vacuole can be scored without a dedicated parasite channel.

BEHIND THE TILES

# Project tools in the Help menu

**Report**
HTML / PDF report of a run: figures, statistics, settings

**Pipeline Graph**
Which run produced which file, as a graph

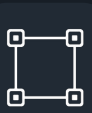
**Batch Runner**
Queue modules with dependencies; validate, then run unattended

**Plate Queue**
Line up plates for Mask / Measure and let them run overnight

**Run History**
Every run, its settings and its outputs

**Distributed Jobs**
Send work to SSH hosts, Slurm or the cloud

**Database Browser**
Browse, search, filter and export project tables



**Data Manager**
What is on disk, how big, and what can go

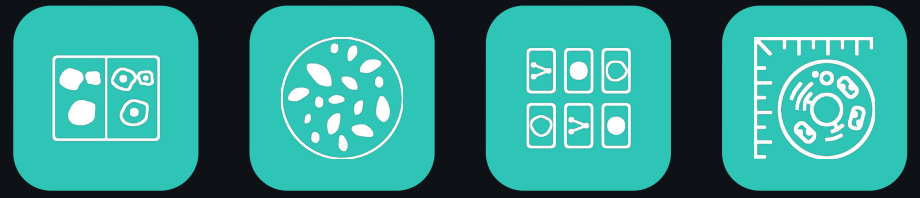
**Project Browser**
All projects on this machine

**Feature Dictionary**
What every measured column means, and its unit

# 05

# Trained models

Models trained for Toxoplasma and host-cell images, shipped through the spaCR Model Zoo. Newest version of each.



MODEL ZOO

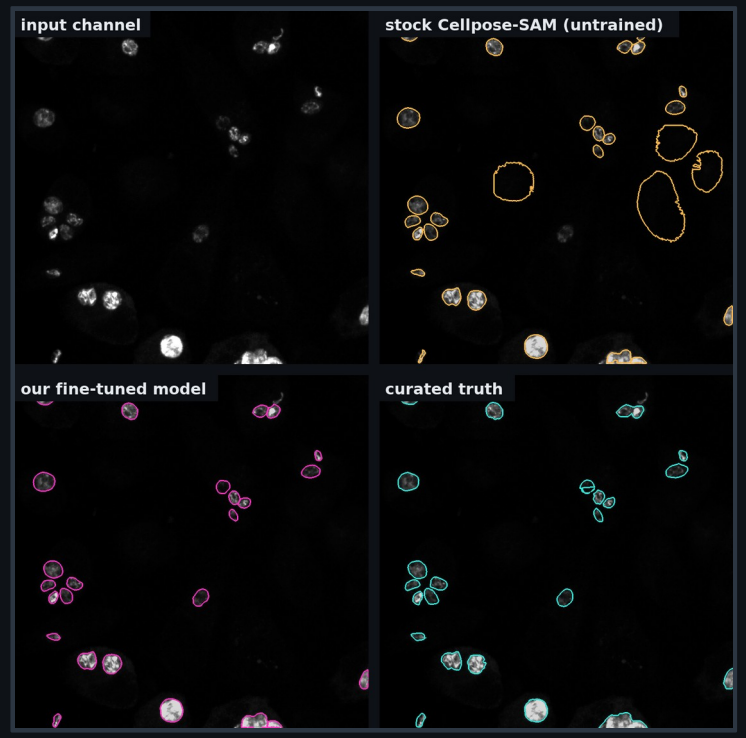
# Eight models, one line in a settings file

| Model | Settings key | Finds | Architecture | Score on unseen images |
|---|---|---|---|---|
| **Toxoplasma PV v4** | `toxoplasma_pv_v4` | Parasite vacuoles | Cellpose-SAM | F1 0.854 (stock 0.765) |
| **Toxoplasma Plaque v4** | `toxoplasma_plaque_v4` | Plaques in wells | Cellpose-SAM | F1 0.815 (v3 0.844) |
| **Plaque Well Detector v2** | `toxoplasma_well_detector_v2` | Wells in plates / figures | YOLO26n | mAP50 0.946 |
| **Toxoplasma from Cell Mask** | `toxoplasma_from_cellmask_v1` | PVs from the cell channel | Cellpose-SAM | F1 0.606 (stock 0.021) |
| **Nuclei from Cell Mask** | `nuclei_from_cellmask_v1` | Nuclei from the cell channel | Cellpose-SAM | F1 0.888 (stock 0.201) |
| **Cell from Hoechst** | `cell_from_hoechst_v1` | Cells from the nuclear channel | Cellpose-SAM | F1 0.870 (stock 0.301) |
| **Toxoplasma from Hoechst** | `toxoplasma_from_hoechst_v1` | PVs from the nuclear channel | Cellpose-SAM | F1 0.569 (stock 0.002) |
| **Live cell v1** | `live_cell_v1` | Unstained cells (phase, DIC) | Cellpose-SAM | F1 0.960 (stock 0.885) |

**How it works** Name a key in a settings file: `pathogen_model: toxoplasma_pv_v4`: and spaCR downloads it the first time it is needed. Anyone can contribute a trained model: it is uploaded to the spaCR Hugging Face and then appears in every user's Model Zoo.

MODEL · TOXOPLASMA_PV_V4

# Toxoplasma PV v4



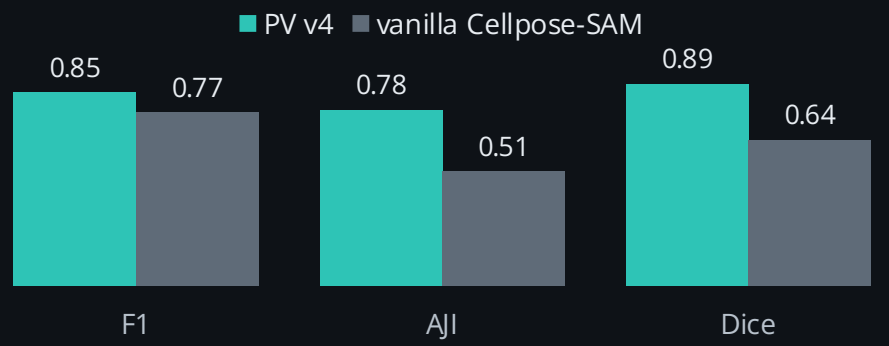

*Unseen test field. Orange = vanilla Cellpose-SAM (cpsam_v2, never trained on our images), magenta = PV v4, cyan = curated truth*

## WHAT THE SCORES MEAN

**F1** share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
**AJI** how well each outline matches; lower if a cell is split in two or two cells merged.
**Dice** pixel overlap between the model's mask and the true mask.
**stock** vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

**Cellpose-SAM (cpsam_v2), round 7**
Outlines Toxoplasma parasitophorous vacuoles in anti-Toxoplasma or DsRed images (RH and ME49).

**11 wells never used for training**




## TRAINED ON

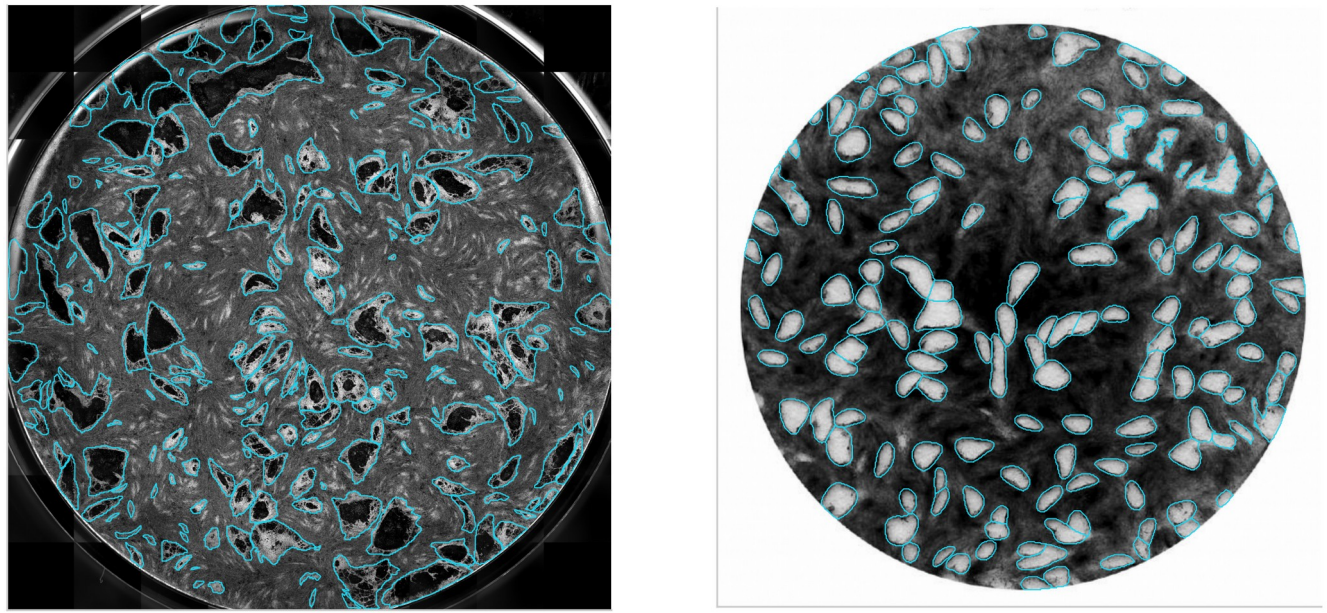
Round 6's 556 curated fields plus 80 from a new plate: 23,340 vacuoles in 502 training fields. 11 wells have never been used for training.

## GOOD TO KNOW

- Ties PV v3 on these wells (F1 0.854 vs 0.860)
- Across five different train / test splits: F1 0.81
- On the new plate it was trained on, PV v3 finds more vacuoles (F1 0.90 vs 0.75): use v3 there
- Traces the vacuoles it finds more tightly than v3

MODEL · TOXOPLASMA_PLAQUE_V4

# Toxoplasma Plaque v4



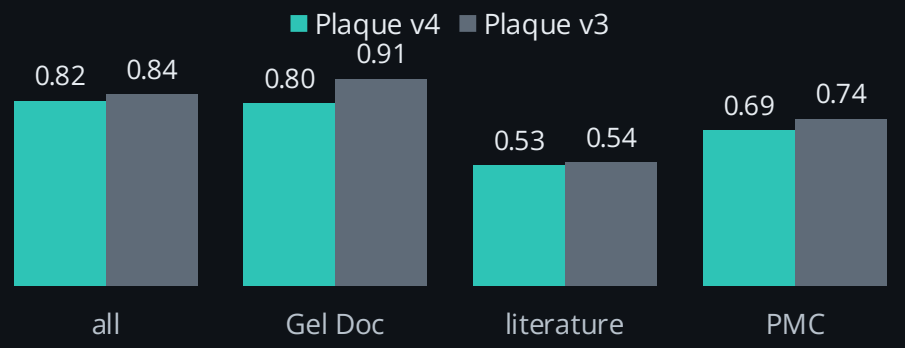
*Curated plaque outlines (cyan): our microscope (left) and a published figure (right)*

## WHAT THE SCORES MEAN

**F1** share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
**AJI** how well each outline matches; lower if a cell is split in two or two cells merged.
**Dice** pixel overlap between the model's mask and the true mask.
**stock** vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

**Cellpose-SAM (cpsam_v2), round 6**
Outlines plaques in crystal-violet wells: from gel imagers, our plates and published figures.

**F1 on 263 unseen fields, by image source**




## TRAINED ON

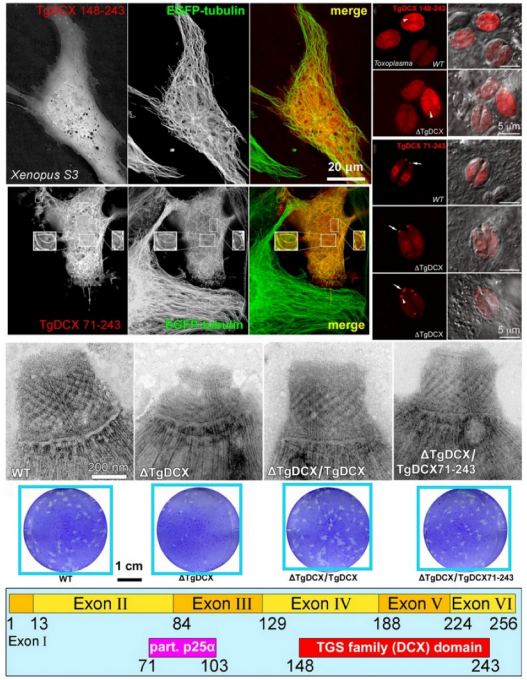
663 curated fields in four domains: published-figure wells, Gel Doc wells, lab plates and literature crops: 32,717 plaques.

## GOOD TO KNOW

- Default since 2026-10-11: built on cpsam_v2
- Plaque v3 scores higher on every source and stays selectable
- Far above v2 on Gel Doc wells (0.80 vs 0.01)
- Balanced precision / recall (0.83 / 0.81)

MODEL · TOXOPLASMA_WELL_DETECTOR_V2

# Plaque Well Detector v2




*Detected wells (cyan) in a published figure; fluorescence and EM panels ignored. Shown with the previous detector*

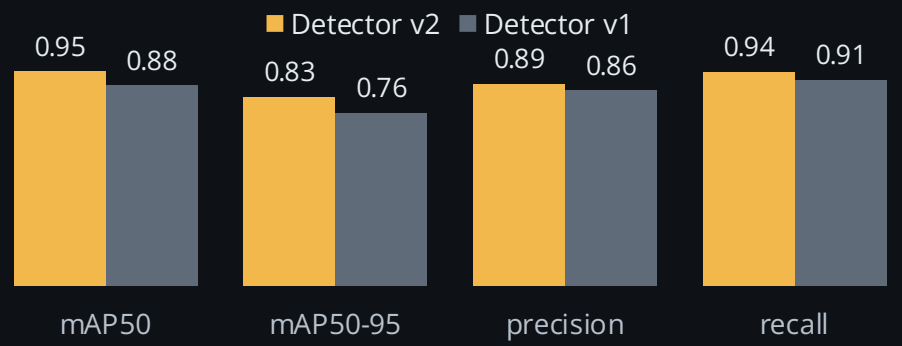
## WHAT THE SCORES MEAN

**precision** of the boxes it draws, the share that really are wells.
**recall** of the real wells, the share it finds.
**mAP50** overall detection score; a box counts when it overlaps the true well by at least half.
**mAP50-95** the same, but also rewards boxes drawn tightly around the well.

## YOLO26n object detector

Draws a box around every plaque well in a plate photo or a published figure: the first step before plaque segmentation.

**129 unseen images, 84 of them with no well**




## TRAINED ON

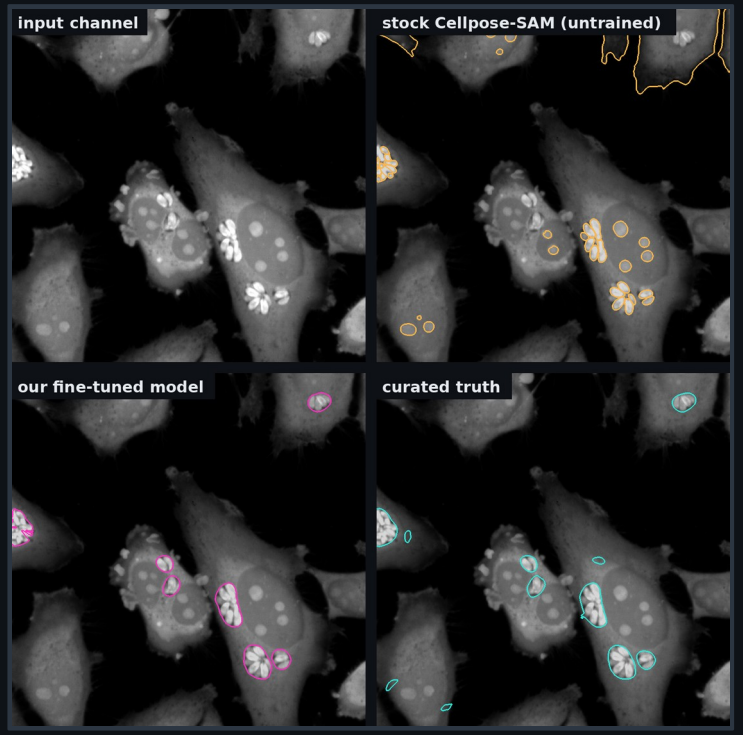
1,070 training images: our plates plus 939 figures from published papers. No paper appears in both training and testing.

## GOOD TO KNOW

- Better than v1 on every score
- False wells on figures with no well: 152 → 49
- Finds wells, not plaques; the well size also gives the mm² scale

MODEL · TOXOPLASMA_FROM_CELLMASK_V1

# Toxoplasma from Cell Mask

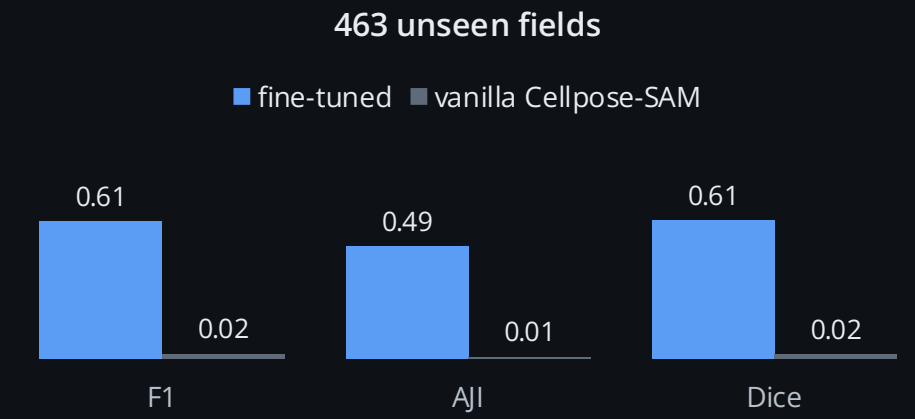



*Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth*

## WHAT THE SCORES MEAN

**F1** share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
**AJI** how well each outline matches; lower if a cell is split in two or two cells merged.
**Dice** pixel overlap between the model's mask and the true mask.
**stock** vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

**Cellpose-SAM (cpsam_v2), cross-channel**
Finds Toxoplasma vacuoles from the host cell-mask channel alone: no parasite stain needed.




## TRAINED ON

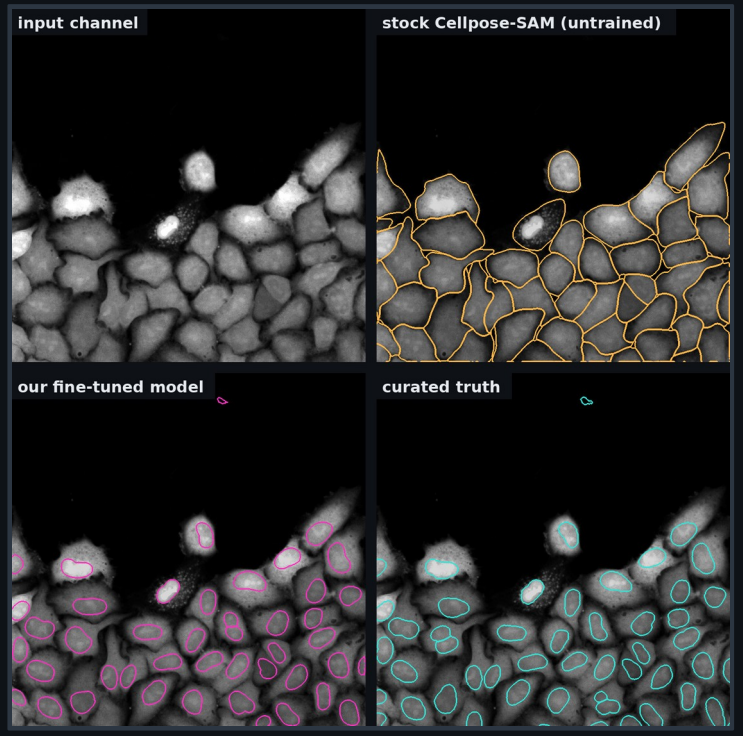
2,567 training fields; tested on 463 fields from other wells. HFF, HeLa and THP-1 hosts.

## GOOD TO KNOW

- Stock Cellpose cannot do this at all
- F1 by host: HeLa 0.71 · HFF 0.56 · THP-1 0.47
- For counting, occupancy and area: not fine morphology

MODEL · NUCLEI_FROM_CELLMASK_V1

# Nuclei from Cell Mask

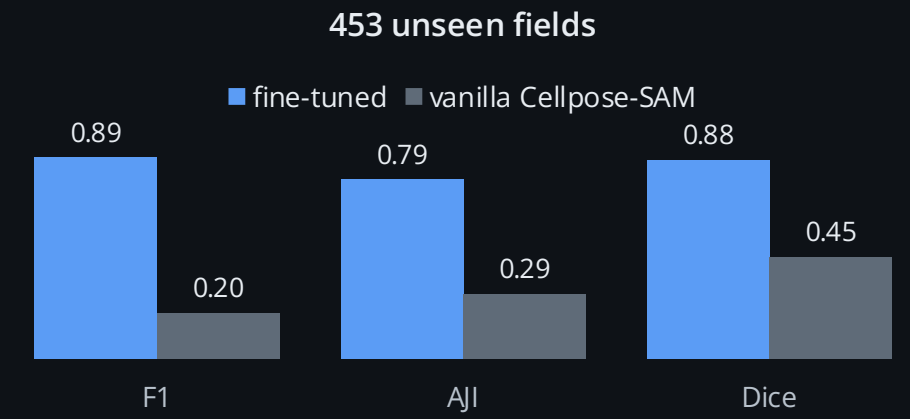



*Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth*

## WHAT THE SCORES MEAN

**F1** share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
**AJI** how well each outline matches; lower if a cell is split in two or two cells merged.
**Dice** pixel overlap between the model's mask and the true mask.
**stock** vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

**Cellpose-SAM (cpsam_v2), cross-channel**
Finds nuclei from the cell-mask channel alone: frees the DAPI / Hoechst channel for another marker.




## TRAINED ON

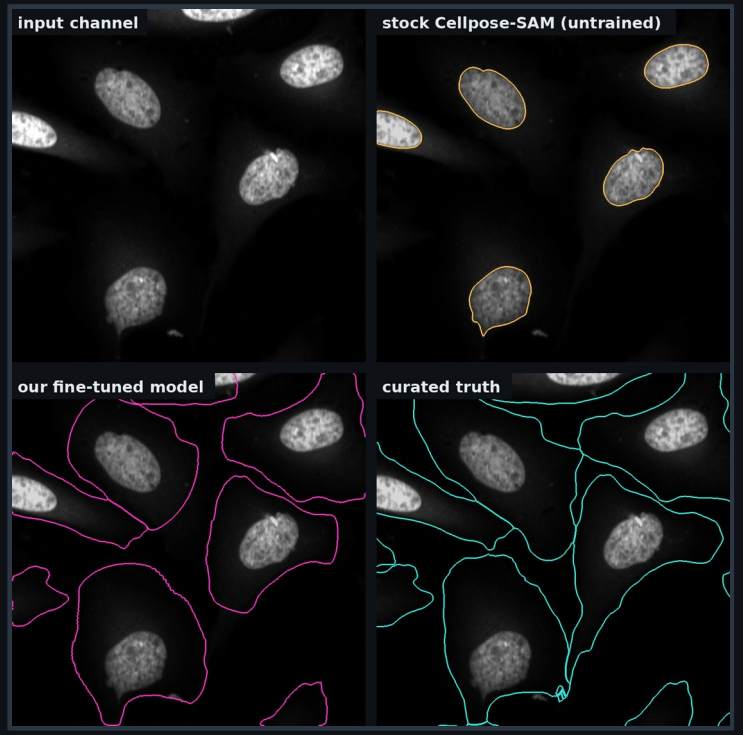
Cell-mask images paired with curated nuclear masks; tested on 453 fields from wells never used for training. HFF, HeLa and THP-1.

## GOOD TO KNOW

- F1 by host: HFF 0.93 · HeLa 0.86 · THP-1 0.86
- Learns nuclei from cell shape: expect worse on unusual or very confluent cells

MODEL · CELL_FROM_HOECHST_V1

# Cell from Hoechst

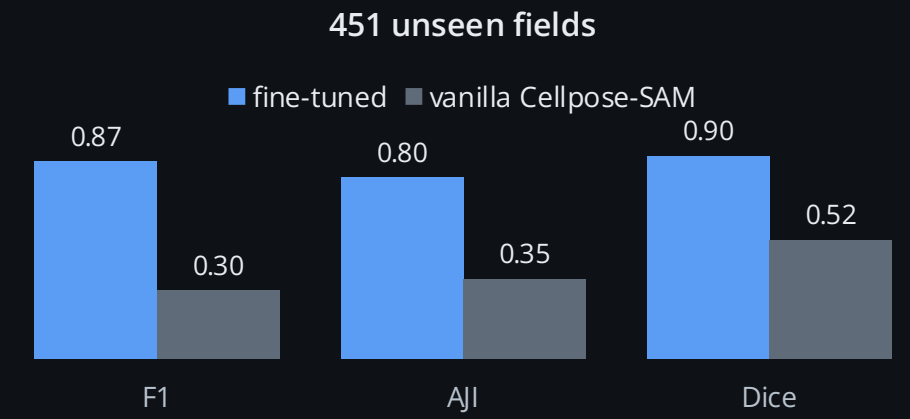



*Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth*

## WHAT THE SCORES MEAN

**F1** share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
**AJI** how well each outline matches; lower if a cell is split in two or two cells merged.
**Dice** pixel overlap between the model's mask and the true mask.
**stock** vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

**Cellpose-SAM (cpsam_v2), cross-channel**
Finds whole host cells from the Hoechst (nuclear) channel alone: the counterpart of nuclei-from-cell-mask.




## TRAINED ON

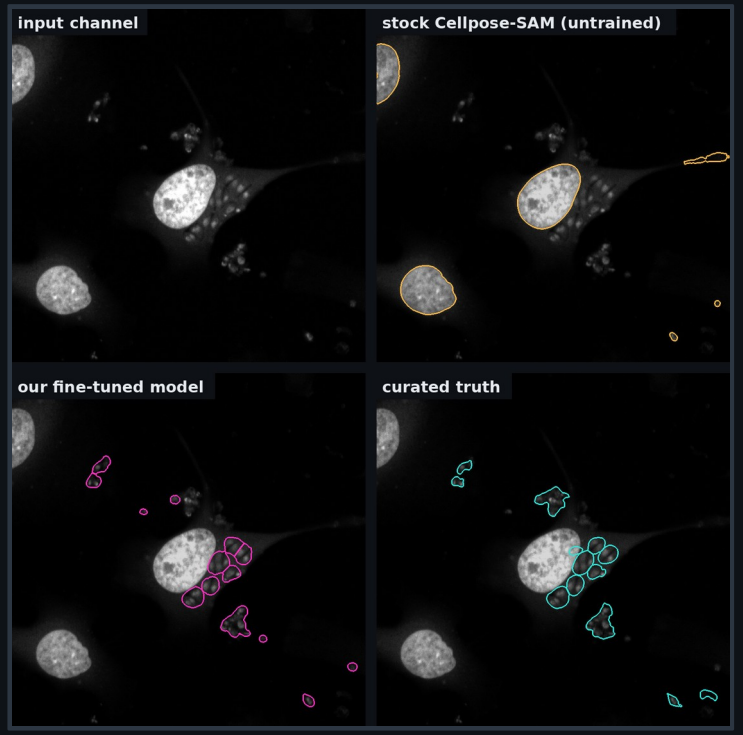
2,578 training fields (237,957 cells); tested on 451 fields (45,098 cells) from other wells.

## GOOD TO KNOW

- Misses some cells but rarely invents one (precision 0.94, recall 0.81)
- F1 0.57 higher than stock

MODEL · TOXOPLASMA_FROM_HOECHST_V1

# Toxoplasma from Hoechst

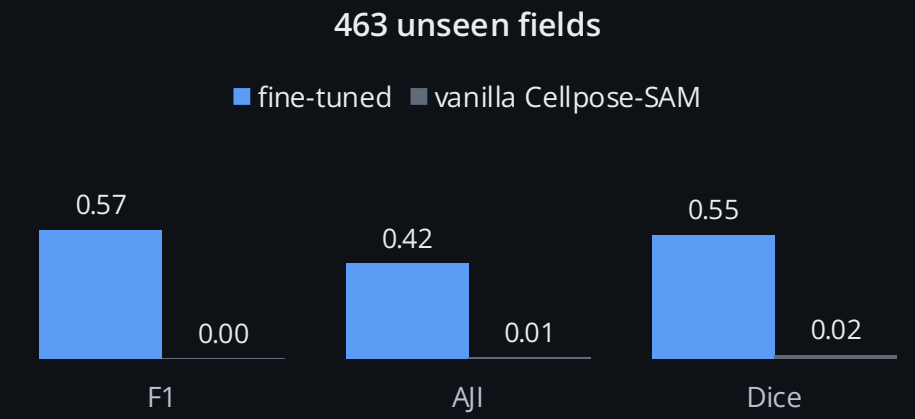



*Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth*

## WHAT THE SCORES MEAN

**F1** share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
**AJI** how well each outline matches; lower if a cell is split in two or two cells merged.
**Dice** pixel overlap between the model's mask and the true mask.
**stock** vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

**Cellpose-SAM (cpsam_v2), cross-channel**
Finds Toxoplasma vacuoles from the Hoechst channel alone: no parasite stain needed.




## TRAINED ON

2,567 training fields; tested on 463 fields from other wells. HFF, HeLa and THP-1 hosts.

## GOOD TO KNOW

- Harder than the cell-mask route (F1 0.57 vs 0.61)
- Stock Cellpose finds essentially nothing
- For counting, occupancy and area

MODEL ZOO

# Other models available in spaCR

## Segmentation

- Cellpose 4: Cellpose-SAM (default) and Cellpose-DINO
- Cellpose 3 models (cyto3, nuclei, ...)
- Cellpose models from bioimage.io
- Models other users have contributed

## Image classifiers (Classify)

- ResNet (default ResNet-50), ResNeXt, Wide ResNet
- EfficientNet, MobileNet, RegNet
- ConvNeXt, VGG, DenseNet
- Vision transformers: ViT, Swin, MaxViT
- All start from networks pre-trained on everyday photos

## Feature classifiers (Classify)

- XGBoost (default), LightGBM, CatBoost
- Random forest and other tree models
- Logistic regression, support vector machine, small neural network

## Embeddings & detection

- Embeddings: ResNet-18 (default), ResNet-50, ConvNeXt, ViT
- YOLO object detection (behind the well detector)

# Get started

```
pip install "spacr[qt]"   &&   spacr
```

| | |
|---|---|
| **Docs & tutorials** | einarolafsson.github.io/spacr |
| **Source & installers** | github.com/EinarOlafsson/spacr |
| **Preprint** | bioRxiv 10.64898/2026.07.08.737057 |
| **Models** | huggingface.co/einarolafsson |
| **Example data** | spacr-download --list |